



Research paper

Open-set Internet of Things intrusion detection via an adaptive few-shot incremental learning framework enhanced with feature augmentation

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ABSTRACT

With the widespread deployment of heterogeneous and resource-constrained devices in real-world Internet of Things applications, detecting previously unseen cyberattacks, referred to as the open-set intrusion detection problem, has become increasingly challenging. This difficulty stems not only from the increased heterogeneity of Internet of Things network traffic, which complicates robust feature extraction under scarce samples, but also from the absence of adaptive and continual learning mechanisms that can dynamically accommodate evolving attack patterns. To address these challenges, this paper proposes a novel feature augmentation and adaptive few-shot incremental learning framework for open-set intrusion detection. Specifically, a feature augmentation network is constructed by combining a dual Residual-enhanced Transformer architecture with three carefully designed and complementary augmentation strategies to enhance discriminative representation learning under limited data conditions. An adaptive incremental classifier is then built to rapidly incorporate new attack categories while preserving global decision consistency. Furthermore, a meta-learning-based pseudo-incremental training strategy is designed to generate synthetic incremental tasks, enabling scalable and robust learning in open-set scenarios. Extensive experiments conducted on four datasets demonstrate that our model consistently outperforms existing baseline and state-of-the-art approaches, achieving an average detection accuracy of over 96% while significantly reducing computational time and memory overhead, highlighting its effectiveness and suitability for resource-constrained environments.

1. Introduction

With the rapid advancement of Internet of Things (IoT), a wide range of terminal devices, including smart home appliances, wearable electronics, and network cameras, have been extensively deployed in practical engineering applications, such as smart homes, intelligent healthcare, and industrial monitoring systems, and are increasingly connected to the Internet. Meanwhile, the development of the IoT has enabled large-scale data exchange and collaborative computing across heterogeneous and resource-constrained devices, significantly expanding the scale, complexity, and operational demands of real-world networked engineering systems (W. et al., 2024). This rapid growth has not only led to unprecedented increases in network traffic volume but has also substantially intensified the frequency, diversity, and sophistication of malicious attacks targeting IoT systems. Such security incidents pose serious threats to device integrity, user privacy, enterprise operations, and even national critical infrastructure (K. et al., 2025; Aguru and Erukala, 2024; Phipps and Nurse, 2026). However, many IoT devices are designed with primary emphasis on functionality,

cost efficiency, and long-term deployment, often at the expense of robust security mechanisms, which renders them particularly vulnerable to attacks. Moreover, the open and highly interconnected nature of IoT architecture further enlarges the attack surface, enabling adversaries to exploit heterogeneous protocols, distributed deployments, and resource-constrained devices. As a result, traditional rule-based or signature-driven security solutions struggle to cope with the scale, diversity, and dynamic evolution of IoT threats, thereby highlighting the necessity of intelligent, data-driven, and adaptive security frameworks based on artificial intelligence for practical IoT security applications (Ren et al., 2025).

In recent years, deep learning-based techniques have been widely validated as effective security solutions for intrusion detection in IoT systems (Ma et al., 2023). In general, intrusion detection systems (IDS) aim to enhance IoT security by continuously monitoring device behaviors and network activities to identify malicious actions and trigger corresponding countermeasures. Despite their demonstrated effectiveness in recognizing known attack patterns, most existing IDS

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approaches are developed under the closed-set learning assumption, which presumes that all attack categories encountered during deployment have already been observed during the training phase (Geng et al., 2020). However, this assumption fundamentally conflicts with the open, dynamic, and continuously evolving nature of real-world IoT environments. In practical deployments, novel and previously unseen attack behaviors inevitably emerge, while conventional classifiers lack the ability to explicitly recognize or reject such unknown threats and therefore tend to misclassify them as known categories. This limitation not only degrades detection reliability but also exposes a fundamental mismatch between closed-set learning and the inherent openness of IoT systems. Given that the occurrence of unknown attacks is unavoidable in IoT scenarios, intrusion detection models must be endowed with open-set recognition capabilities to explicitly identify and handle unknown threats, thereby enabling a principled transition from closed-set learning to open-set IoT intrusion detection (Wu et al., 2024; Sahu et al., 2024; Wang et al., 2026).

In open-set intrusion detection, unseen-category data refer to samples that are absent from the training set and whose feature distributions differ from those of known attacks, representing unknown IoT threats such as novel or variant attacks, as well as anomalies induced by device heterogeneity and environmental variations. Consequently, the core objective of open-set intrusion detection is inherently dual in nature: accurately classifying known attack categories while effectively identifying and handling previously unseen threats. Although several preliminary studies, including OpenMax (Bendale and Boulton, 2016; Zhang et al., 2024), extreme value machine-based models (EVM) (Zhao and Cheong, 2026), prototype compositional networks (PCN) (Yang et al., 2022), and prototype mixing adversarial learning (PMAL) (Lu et al., 2022), have demonstrated the feasibility of open-set recognition, their direct application to IoT intrusion detection scenarios remain constrained by domain-specific challenges. Specifically, existing open-set detection methods in IoT security applications face two fundamental technical limitations. First, unknown attack samples are extremely scarce in real-world IoT environments, providing insufficient data support for learning robust and discriminative representations (Pan et al., 2023). This challenge hinders the performance of traditional models such as CNN, LSTM, and Transformer-only architecture. CNNs capture local patterns but struggle with long-range dependencies, LSTMs face limitations with high-dimensional and heterogeneous features, and Transformer-only models, while excellent at global modeling, often lack fine-grained local feature extraction. Pure ViT has global modeling capability but tends to produce unstable representations under few-shot conditions, leading to noisy representations. Second, most existing approaches rely on static training paradigms and lack dynamic response mechanisms to accommodate continuously evolving attack patterns, thereby failing to support incremental learning for emerging threats (He et al., 2026). As a result, detection models often suffer from rapid performance degradation, which contradicts the essential requirements of IoT systems for long-term adaptability and real-time defensive capability. Therefore, effectively securing open and evolving IoT environments necessitates an intrusion detection framework that jointly integrates high discriminative feature learning with adaptive incremental learning capabilities.

To address the challenge of learning discriminative representations from scarce and heterogeneous IoT data, feature augmentation is seen as a promising strategy for improving intrusion detection model generalization. By enriching feature representations beyond the original training distribution, feature-level augmentation helps compensate for the lack of unknown attack samples and boosts robustness to unseen threats. However, existing methods are typically designed for closed-set or homogeneous data and struggle in open, dynamic IoT environments. They often fail to capture cross-device correlations, model long-range dependencies among heterogeneous IoT features, or preserve discriminative information under complex conditions, leading to redundant or noisy representations that hinder detection of unknown attacks. To

overcome these limitations, this work introduces a Transformer-based feature augmentation framework that uses attention-driven representation learning to model global contextual dependencies and cross-device relationships. By incorporating global-local contextual awareness and attention-based feature prioritization, it highlights salient patterns while suppressing irrelevant noise, enhancing the model's ability to capture diverse attack features and improve robustness against previously unseen attack types for open-set intrusion detection.

Beyond the challenge of feature representation learning, open-set IoT intrusion detection must also address the continuous evolution of attack patterns after deployment, which renders static detection models inadequate for long-term security assurance. In real-world IoT environments, novel attacks do not emerge simultaneously but appear progressively over time, requiring intrusion detection systems to incrementally incorporate new knowledge while preserving previously learned behaviors. Although several studies have attempted to introduce adaptive mechanisms into open-set detection, such as the contextual evolutionary classifier (CEC) (Zhang et al., 2021), which propagates contextual information through graph-based models to establish semantic correlations between known and emerging categories, existing approaches remain insufficient for practical incremental intrusion detection. In particular, the extreme scarcity of emerging attack samples severely constrains the learning of discriminative representations for fine-grained intrusion characterization, as merely enhancing the transferability of feature extraction networks is inadequate to overcome the low-data regime inherent in unknown attack detection. Moreover, the progressive updating of category prototypes during incremental learning may induce representation drift between fixed feature embeddings and dynamically evolving classifier prototypes, especially in open-set scenarios where new knowledge is continuously integrated. These limitations underscore the necessity of a more adaptive and data-efficient incremental learning framework capable of supporting reliable and sustained intrusion detection in dynamically evolving IoT environments.

To tackle the challenges of open-set IoT attack detection, we propose feature augmentation and adaptive few-shot incremental learning framework (FAN-AFSIL), a framework that combines feature augmentation and adaptive few-shot incremental learning in three collaborative stages: FAN pretraining, incremental learning, and open-set recognition. In the FAN pretraining stage (Stage I), we address the issue of learning discriminative representations from scarce, heterogeneous IoT data by designing a feature augmentation network (FAN) based on a dual residual Transformer architecture, which uses group-attention and global-local feature extraction to enhance critical attack feature representation. In the incremental learning stage (Stage II), we introduce an adaptive incremental classifier (AIC) for rapid few-shot adaptation and alignment with evolving feature distributions. The AIC incorporates a hybrid relational projection (HRP) network to model contextual relationships among category prototypes and a pseudo-episodic training (PET) strategy to reduce representation drift and improve stability during incremental updates. In the open-set recognition stage (Stage III), a pseudo-episode selection (PES) mechanism is used to construct pseudo-incremental tasks with mixup-based augmentation, enhancing the model's ability to identify unknown attacks and supporting robust incremental learning. Overall, FAN-AFSIL addresses both the feature complexity of IoT heterogeneity and the dynamic evolution of attack patterns, enabling efficient and accurate open-set intrusion detection in real-world IoT environments. Unlike traditional approaches that focus on either local feature extraction or global modeling, FAN-AFSIL integrates both paradigms within a unified framework, enabling the model to effectively capture fine-grained details and long-range dependencies. Rather than simply combining existing components, FAN-AFSIL introduces an innovative architecture that brings together local feature extraction and global modeling. Each module is carefully integrated to address the challenges of open-set recognition, few-shot learning, and incremental adaptation. By decoupling feature extraction from task

adaptation, FAN-AFSIL not only enhances expressiveness and generalization but also ensures dynamic adaptability in complex, evolving IoT environments. The main contributions of this work are summarized as follows.

- We develop a feature augmentation network integrating a global-local feature extraction module (GLFEM) and a group-attention dual residual module, which effectively extracts transferable and discriminative IoT intrusion features under data-scarce and heterogeneous open-set conditions. This architecture enhances the model's capability to characterize unknown attacks through multi-granularity representation learning.
- To address prototype drift and adaptation challenges in incremental learning, we design an adaptive incremental classifier (AIC) based on meta-training. Specifically, a prototype self-projection network dynamically updates the prototypes of both known and novel categories, while a cross-projection network refines feature representations to maintain consistency with evolving category prototypes.
- We propose a pseudo-episodic training (PET) scheme to enable data-efficient and stable few-shot incremental learning in open-set environments. By generating diverse pseudo-incremental tasks via a pseudo-episode selection (PES) strategy and leveraging meta-learning, the proposed scheme endows the hybrid relational projection (HRP) network with continual adaptation capability across evolving attack scenarios.
- Extensive experiments conducted on four representative IoT intrusion detection datasets demonstrate that FAN-AFSIL consistently outperforms existing state-of-the-art approaches. In addition to improved detection accuracy, the proposed framework achieves superior memory efficiency and reduced computational overhead, highlighting its practicality for resource-constrained IoT environments.

The organization of this paper is as follows: Section 1 introduces the research objectives and the rationale behind our model; Section 2 presents a literature review related to this work. Section 3 details the specific implementation of FAN-AFSIL model. In Section 4, we establish a series of experiments to validate the performance of FAN-AFSIL model. Finally, Section 5 summarizes the advantages of this study and outlines the future.

2. Related work

2.1. Deep learning-based intrusion detection in IoT

With the rapid expansion of Internet of Things (IoT) deployments, intrusion detection systems (IDS) have become a critical security component for protecting IoT infrastructures against diverse cyber threats. In recent years, deep learning-based approaches have been extensively investigated for IoT intrusion detection, owing to their strong capability in automatic feature extraction and complex pattern modeling from large-scale network traffic data.

Ding et al. (2024) proposed a GAN-based data augmentation method for IoT intrusion detection, in which a multi-generator architecture is employed to generate diverse attack samples, and the classifier is optimized to enhance the performance of both the generator and discriminator. By computing the cosine similarity between generated samples and original samples as well as among generated samples, a generator loss function is defined to improve sample quality and reduce category overlap. Experimental results demonstrate that this method achieves favorable performance on relevant IoT datasets. Nakip and Gelenbe (2024) proposed a self-supervised intrusion detection (SSID) framework that enables a fully online deep learning-based IDS without human intervention or offline training. This approach is based on the decision mechanism of self-associative deep random neural

networks and online confidence estimation, allowing the model to adapt to dynamic changes in network traffic while eliminating the need for offline data collection. X. Yang et al. (2025) proposed an efficient and lightweight dynamic open-set intrusion detection scheme for industrial IoT (IIoT). The proposed method consists of three stages, including known attack classification, unknown attack identification, and dynamic update detection, thereby enabling effective detection of both known and unknown attacks. Yan et al. (2025) proposed a personalized federated meta-learning intrusion detection method, namely PerFLID, which supports multiple participants in customizing detection models according to local data features and achieving local adaptability. This method aims to optimize client-side local models and adopts a gradient similarity-based client selection strategy to enhance effective contributions to the global model, thereby accelerating model convergence.

Duan et al. (2024) proposed an interaction-centric network anomaly detection framework based on continuous-time temporal graph (CTG) neural networks. By modeling the interaction processes among network entities as CTG evolution, the framework effectively represents new node access behaviors. Experimental results show that the proposed method exhibits superior detection performance for new access events and low-frequency anomalous behaviors across multiple datasets. Lu et al. (2025) proposed a multi-objective discrete evolutionary optimization model, termed MODEO-CNN, which simultaneously performs neural architecture search and hyperparameter optimization to automatically construct lightweight CNN models for IIoT intrusion detection. The method employs hybrid variable-length encoding to characterize network architectures and key hyperparameters and utilizes multi-objective evolutionary search to obtain Pareto-optimal models. Experimental results indicate that MODEO-CNN outperforms existing methods in terms of both detection performance and computational efficiency on multiple IIoT datasets. Hu et al. (2025) proposed a federated learning-based intrusion detection framework for software-defined networking (SDN)-enabled industrial IoT, named SFLNID. To address non-IID data distribution and data imbalance issues, the framework incorporates focal loss and a joint optimization mechanism based on Wasserstein distance. In addition, dynamic gradient clipping and adaptive differential privacy strategies are integrated to enhance training efficiency and detection performance while preserving data privacy. Wu et al. (2025) proposed an attack-aware divide-and-conquer Transformer model for IoT intrusion detection. It achieves high accuracy in anomaly detection and multi-class attack classification across multiple IoT and IIoT datasets, significantly reducing false positive rates. Overall, Tracer improves performance compared with existing methods, demonstrating its potential as a general-purpose IIoT intrusion detection solution. Zeng et al. (2025) proposed a causal deep learning-based intrusion detection system for IoT that can perform attack classification. Experimental results demonstrate that it achieves high classification accuracy for attack types across multiple representative IoT datasets.

In summary, deep learning-based intrusion detection models for the IoTs have achieved notable advances. However, most existing approaches are developed under the closed-set assumption, where attack categories observed during testing are assumed to be consistent with those seen in training. In real-world IoT environments with continuously emerging unknown or variant attacks, this assumption significantly limits detection performance and generalization capability. Consequently, effectively identifying unknown attacks while preserving high performance on known attacks remains a critical challenge. To address this issue, Section 2.2 reviews recent advances in open-set intrusion detection for IoT environments.

2.2. Open-set intrusion detection

As analyzed in Section 2.1, the closed-set assumption underlying most existing detection methods restrict their generalization capability in real-world environments with continuously emerging attacks.

Accordingly, this section reviews recent advances in open-set intrusion detection, which aims to maintain high detection performance for known attacks while effectively identifying unknown categories in IoT scenarios. Existing open-set intrusion detection methods can be broadly categorized into threshold-based approaches and prototype-based representation learning methods. Threshold-based methods, such as OpenMax (Bendale and Boulton, 2016; Zhang et al., 2024) and entropy-based models (EVM) (Yu et al., 2023), estimate the likelihood of unknown samples by modeling decision boundaries or output uncertainty. Although these methods have demonstrated effectiveness in generic open-set recognition tasks, they typically rely on well-separated feature distributions and stable category boundaries, which are difficult to guarantee in highly variable and noisy IoT traffic. Prototype-based methods, including prototype compositional networks (PCN) (Yang et al., 2022) and prototype mixing adversarial learning (PMAL) (Lu et al., 2022), aim to enhance unknown-category detection by learning representative prototypes and encouraging separability. However, their performance may degrade in IoT environments due to severe category imbalance, evolving attack patterns, and device heterogeneity.

In addition to the above approaches, a variety of other open-set intrusion detection methods have been proposed from different perspectives. For example, Souza et al. (2025) introduced an improved one-category energy flow classifier for open-set network intrusion detection, achieving both accurate known-attack classification and effective unknown-attack recognition with low computational complexity. Yu et al. (2023) proposed a deep reinforcement learning-based open-set IDS for IIoT, which combines deep Q-networks with conditional variational autoencoders to distinguish unknown attacks using reconstruction errors. Several studies have explored memory-based and isolation-based strategies, such as multi-scale feature memory banks (Shao et al., 2025) and isolation tree-driven open-set detection frameworks (Z. Zhao et al., 2024), to improve robustness against unknown attacks. Moreover, unsupervised domain adaptation (Gao et al., 2024), multimodal representation learning (Bai et al., 2025), state-space-based feature modeling (Z. Wang et al., 2025), and enhanced zero-day attack detection frameworks (Arthur et al., 2025) have also been investigated to improve open-set detection performance under diverse IoT scenarios.

Despite these advances, most existing open-set intrusion detection models are designed under static learning assumptions and lack the ability to continuously adapt to newly emerging attack categories. When novel attacks appear progressively, retraining models from scratch is often impractical, especially in resource-constrained IoT environments. Therefore, enabling efficient and stable incremental learning for open-set intrusion detection remains an open challenge, which will be discussed in the next section.

2.3. Incremental learning

As highlighted in Section 2.2, the ability of intelligent systems to continuously learn from evolving data without catastrophic forgetting is crucial. Incremental learning has emerged as a key paradigm for supporting continuous model adaptation. Existing incremental learning methods can be broadly classified into three strategies: regularization-based approaches that limit parameter updates to preserve previously learned knowledge, replay-based mechanisms that retain historical information through data or feature rehearsal, and dynamic architectural strategies that expand or reorganize network structures to accommodate new tasks.

Regularization-based models aim to mitigate the forgetting of knowledge from previous tasks by constraining the parameters associated with historical tasks when learning new categories. For instance, Zhang et al. (2025) proposed an attention-based multimodal weighted canonical correlation analysis mechanism, which integrates replay and regularization methods to derive a unified process detection model. Y. Shi et al. (2021) introduced a bit-level information preservation (BLIP) mechanism, which leveraged bit-level parameters to retain

the information. S. Wang et al. (2025) proposed a novel network training algorithm, which employed singular value decomposition to approximate the null space of the non-centered covariance matrix. Replay-based models address catastrophic forgetting by preserving historical knowledge through either storing subsets of old samples. For instance, J. Yang et al. (2025) designed a data compression mechanism integrated with a determinantal point process (DPP) to regulate sample compression quality, thereby achieving efficient memory replay. Gao et al. (2022) proposed a relation-guided deep feature consolidation for incremental learning (R-DFCIL) model, which improved generative replay performance. Toldo and Ozay (2022) estimated semantic shifts across incremental stages, then inferred current representations of old-category prototypes. Zhu et al. (2025) combined prototype augmentation with self-supervised learning, maintaining prototypes for historical categories and enhancing robustness. Zhu et al. (2022) introduced a self-sustaining representation expansion scheme, where dynamically reorganized network structures incorporate branches to refine incremental learning for new categories.

Existing incremental learning models typically require sufficient training samples to perform well. However, in IoT scenarios, new categories often have very limited samples, leading to significant performance degradation in conventional algorithms. To tackle this issue, few-shot incremental learning (FSIL) has emerged as a key research focus, aiming to maintain model plasticity while preserving stability. G. Shi et al. (2021) pioneered an optimization strategy that identifies flat local minima of the objective function. By fine-tuning model parameters, they enhance adaptability to novel scenarios. Li et al. (2025) designed a distribution estimation module to preserve the feature distribution of previously learned categories and introduced a data replay module to randomly sample features from earlier categories, thereby mitigating catastrophic forgetting. Chi et al. (2022) introduced a meta-learning optimization paradigm, where incrementally arriving tasks were simulated by episodically sampling. Xu et al. (2023) built the multi-feature space similarity supplementation model, which constructed task-specific feature spaces via a boundary-sensitive sample attention mechanism. H. Zhao et al. (2024) developed a multi-granularity slow-fast learning strategy, while fast-learning modules adapt to new categories. Zhou et al. (2022) created forward-compatible adaptive classification training (FACT), which compressed base-category embeddings.

Despite recent advancements in incremental learning models, two critical challenges remain insufficiently addressed in IoT attack detection: First, the extreme scarcity of IoT attack scenarios severely limits the model capacity to capture fine-grained discriminative features, even when transferability of feature extraction models is enhanced through existing techniques. Second, the continuous adaptation of old and new category prototypes during task updates may induce representation-classifier mismatch. Furthermore, the persistent scarcity of novel samples and the lack of sufficient data to support classifier expansion exacerbate these issues, creating a compounded degradation in model plasticity and stability over long-term incremental updates. To address these challenges, we propose a FAN-AFSIL framework for IoT intrusion detection, which integrates feature augmentation and adaptive incremental learning for few-shot scenarios. Specifically, the framework incorporates a feature augmentation network to improve model generalization by effectively mining latent discriminative information from sparse samples. In parallel, we introduce an adaptive incremental classifier strategy that dynamically adjusts classifier weights and sample features across different incremental tasks, thereby enabling efficient detection of evolving IoT intrusion behaviors in open-set environments.

3. FAN-AFSIL model

3.1. Problem definition and overall architecture

3.1.1. Problem definition

Few-shot incremental learning (FSIL) aims to train models incrementally from a continuous stream of data. The FSIL setting typically

includes a sequence of labeled training datasets $D^1, D^2, \dots, D^t$, where each dataset satisfies a specific task constraint $D^i = \{I^i, C^i\}$, and t denotes the task index set. Let I^i and C^i represent the samples and corresponding label space of the i task, respectively. In the i th task, only a limited labeled sample D^i are available, subject to the constraint $C^i \cap C^j = \emptyset (i \neq j)$ (the labeled samples are significantly smaller than the size of the category space). In the base task ($t = 1$), the training set D^1 contains N_b categories with a sufficient sample (base dataset). In the incremental task ($t \geq 2$), each category has only a few samples, and the incremental task data is typically described as an N_n -way K -shot setup, where the dataset contains K categories, each with N_n samples (incremental dataset). When learning the i th incremental task, only the current task dataset is available. Moreover, the model objective is not only to learn new knowledge from the current task but also to preserve recognition performance on previously learned tasks. During the testing phase, the model is evaluated on the entire category set $C = \bigcup_{i=1}^t C^i$. In addition, FSIL baseline models are trained solely on the base task and directly applied to inference across all tasks. Specifically, a feature extraction network F^* is trained on the base dataset D^1 by minimizing the standard cross-entropy loss function. Furthermore, a prototype-based nearest category mean (NCM) classifier is employed to perform IoT attack detection. Specifically, all training samples I are mapped into the feature space of the network F^* to obtain their feature representations $x^* = F^*(I)$. The initialization weights of the NCM classifier are denoted as W^* . Euclidean distance $\text{dist}(\cdot, \cdot)$ is employed to measure the semantic discrepancy between a sample feature and its corresponding category prototype W^* . The loss function is formally defined as:

$$f(x^*, W^*) = \frac{e^{-\text{dist}(x^*, W_C^*)}}{\sum_{W_j^* \in W^*} e^{-\text{dist}(x^*, W_j^*)}}, \quad (1)$$

$$L = L_{CE}(f(x^*, W^*), c) \quad (2)$$

After training, a prototype is stored for each category in the base dataset as the updated weight of the NCM classifier. Each category prototype is computed using the mean vector of its corresponding feature representations and can be formulated as:

$$W_C^{*base} = \frac{1}{n_c} \sum_j [y_j = c] x_j^*, \quad (3)$$

In Eq. (3), c represents the category, and n_c denotes the number of training samples for the c th category. If $y_j = c$ is true, then $[y_j = c] = 1$ holds; otherwise, it is zero.

The updated weights of the NCM classifier are represented as:

$$W_{base}^* = \{W_1^{*base}, W_2^{*base}, \dots, W_{N_b}^{*base}\} \in \mathbb{R}^{N_b \times d} \quad (4)$$

In incremental tasks, the feature extraction network F^* is frozen without any fine-tuning on new categories. With the assistance of F^* , the prototypes of the new categories $W_{novel}^* = \{W_1^{*novel}, W_2^{*novel}, \dots, W_{N_n}^{*novel}\} \in \mathbb{R}^{N_n \times d}$ are computed. Accordingly, the classifier's weight matrix is updated as $W^* = [W_{base}^*; W_{novel}^*]$. Specifically, we utilize the NCM classifier to perform classification on both old and new task test features $X^* = [X_{base}^*; X_{novel}^*]$.

A sample is assigned to the category in W^* whose prototype has the minimum Euclidean distance to the feature X^* .

3.1.2. Overall architecture

The overall architecture of the proposed FAN-AFSIL model is illustrated in Fig. 1. Considering the severe imbalance between the number of old categories and new attack instances, the model decouples the learning of the feature extraction network and the classifier. It mainly consists of three stages: (i) the feature augmentation network (FAN) pretraining stage, (ii) the incremental learning stage, and (iii) the open-set recognition stage. The FAN-AFSIL framework learns and adapts unseen-category data via a unified three-stage pipeline. First, to enable processing by the FAN framework, raw network traffic data are

first transformed into image-like time-frequency representations using the Short-time Fourier transform (STFT). This transformation maps time-domain signals to the frequency domain, allowing the model to capture both temporal and spectral features of the network traffic. Specifically, after preprocessing the raw network data, STFT is applied to convert the time-domain signals into frequency-domain representations, which are then used to generate time-frequency spectrograms. These spectrograms are subsequently fed into the CNN/Transformer-based FAN framework for feature extraction. Moreover, FAN network is pre-trained on base categories to extract discriminative global-local representations and establish category prototypes (W_{base}^*). Second, prototype self-projection (PSP) and feature cross-projection (FCP) maintain representation consistency and integrate emerging category prototypes into the shared embedding space, enabling stable incremental updates without disrupting decision boundaries. Third, unseen samples are simulated through category partitioning and pseudo-incremental tasks generated by the PES strategy; these pseudo-tasks meta-train the adaptation module to generalize beyond known categories. Together, these stages allow the model to effectively process unseen data and sustain robust open-set recognition performance.

FAN pretraining stage: We design a feature augmentation network (FAN) based on a dual Residual-enhanced Transformer (Res-Transformer), which integrates a global-local feature extraction module with a grouped-attention dual-residual module. This design enables the network to capture both holistic contextual information and fine-grained discriminative patterns, thereby improving feature representation under IoT environments. As a result, the pretrained FAN enhances the model's ability to perceive and generalize to previously unseen IoT attack samples. The dual Res-Transformer is pretrained on the base dataset to provide robust feature embeddings for subsequent incremental learning stages. Upon completion of pretraining, a prototype W_{base}^* is stored for each base category.

Incremental learning stage: In this stage, the FAN is frozen to effectively mitigate overfitting on new tasks and alleviate catastrophic forgetting of previous tasks. Meanwhile, a hybrid relational projection network (HRP) is trained through meta-learning. Specifically, pseudo-incremental data are generated using a pseudo-episode selection (PES) strategy, including a pseudo incremental support set (PIS) and a pseudo incremental query set (PIQ). It should be noted that freezing FAN does not mean that the model loses its ability to adapt to new attack patterns. In our framework, adaptation is achieved through relational reconstruction rather than feature extractor updating. Specifically, PSP updates the prototype structure by modeling correlations between old and new categories, while FCP refines feature-to-prototype alignment under the updated relational space. As a result, even with a fixed backbone, the semantic interpretation of new samples can still be dynamically adjusted to accommodate evolving attack distributions. Additionally, a base query set (BQ) is also sampled from the base dataset and incorporated into the HRP training process, which enables dynamic adaptation to new tasks without forgetting previous knowledge. The PIS, PIQ, and BQ are passed through the FAN to obtain representations W_{novel} , X_{novel} , and X_{base} , respectively. In parallel, W_{base} and W_{novel} are fed into the prototype self-projection network (PSP) to generate updated prototypes W_{new}^* . Furthermore, W_{base}^* , W_{novel}^* , X_{novel} , and X_{base} are input to the feature cross-projection network (FCP) to produce the updated representation X_{new} . Finally, the representations X_{new} are classified using the NCM classifier and trained with the standard cross-entropy loss, thereby enabling open-set IoT intrusion detection. During the training of pseudo-incremental tasks, if the AIC can effectively calibrate both the prototypes and feature representations of old and new categories, it is expected to retain its effectiveness.

Open-set recognition stage: After the FAN-AFSIL training, the model acquires the ability for continual learning. In this stage, both the FAN and HRP modules are frozen and deployed to real-world incremental tasks. When a new category of intrusion is detected, the prototype representation W_{novel}^* of the new attack is obtained through the FAN

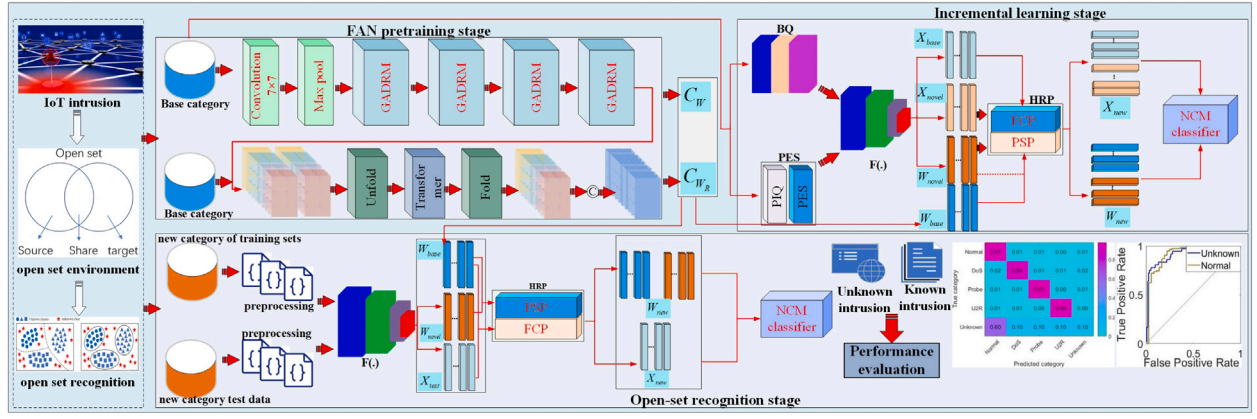


Fig. 1. FAN-AFSIL.

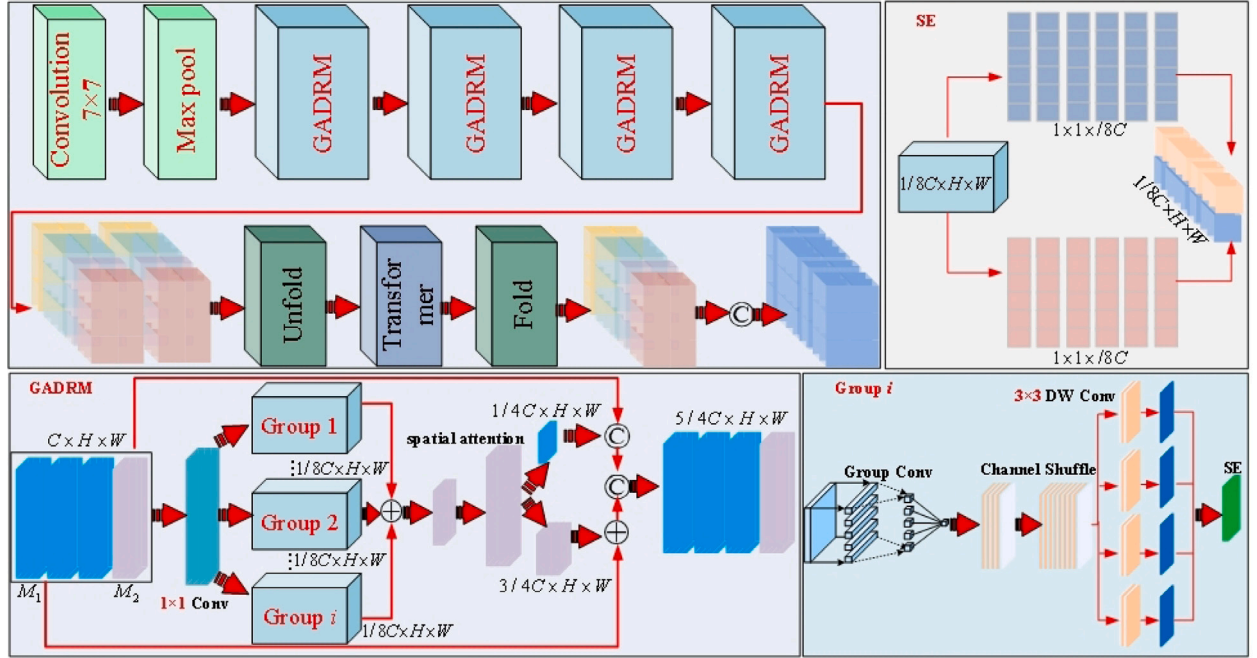


Fig. 2. Dual Res-Transformer.

(which, together with the base intrusion prototype W_{base} , is fed into the PSP module to yield an updated prototype W_{new}). When an attack sample needs to be recognized, the attack representation X_{test} and its prototype W_{new} , along with the base prototypes W_{base} , are input into the FCP to produce the updated representation X_{new} . Finally, classification is performed through the NCM classifier. The updated representation W_{new} is then merged into W_{base} as the old prototype, thus forming $W_{base} = [W_{base}; W_{new}]$.

3.1.3. FAN framework construction

A key challenge in detecting unknown IoT intrusions under the few-shot incremental learning (FSIL) setting is to construct a robust feature extractor that can effectively represent attack patterns with limited labeled data. The high heterogeneity of IoT devices and attack behaviors further increases the difficulty of generalizing unseen attacks. From a representation learning perspective, this task requires the model to capture local discriminative patterns, global dependencies across devices, and maintain stable optimization under data scarcity. However, single-structure models are theoretically limited in meeting these requirements simultaneously. CNN-based models rely on local receptive fields and struggle to model long-range dependencies.

Transformer-based models capture global relationships but lack strong local inductive bias, which may lead to unstable representations in few-shot scenarios. Pure ViT models often suffer from overfitting and noisy feature representations when training data is limited. Therefore, relying on a single backbone is insufficient for complex IoT intrusion modeling.

Therefore, we propose a FAN model called the dual Res-Transformer (Fig. 2), which enhances the model's feature extraction capability by incorporating two distinct feature augmentation strategies. Specifically, we design a grouped attention dual Residual module (GADRM) that employs a dual-residual architecture to enable efficient feature fusion, thereby enhancing the model ability to capture critical IoT attack patterns. Additionally, a global-local feature extraction module (GLFEM) is constructed to obtain global features with high-level semantic information, further improving the model capacity for representation of IoT intrusions. This unified architecture enables multi-granularity feature learning and improves representation capacity. It provides a theoretically justified and task-aligned solution that overcomes the limitations of single-structure models in open-set IoT intrusion detection under few-shot conditions.

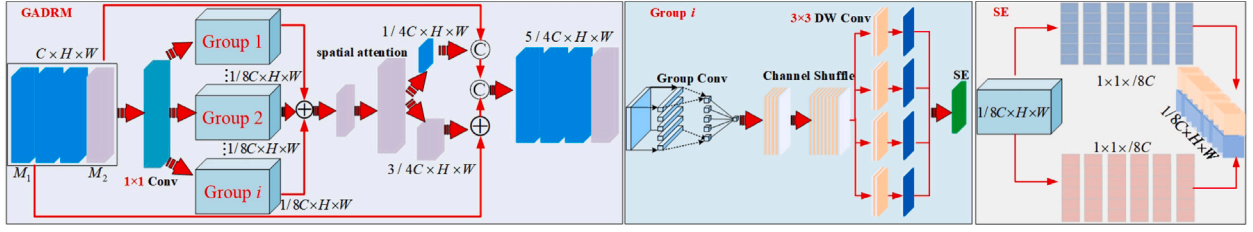


Fig. 3. GADRM module.

3.1.4. GADRM strategy construction

As illustrated in Fig. 3, our proposed GADRM integrates two distinct residual connections to enable iterative feature refinement and enhance the module's ability to capture region-specific features of IoT attacks. The channel shuffle operation first facilitates the fusion of inter-channel information from feature maps produced by group convolution, promoting better feature interaction. The fused feature map is then partitioned into four subgroups, each undergoing distinct transformations to extract complementary features. Depth-wise separable convolution is employed to improve intra-group feature extraction and discriminability. Additionally, the module incorporates squeeze-and-excitation (SE) operations to adaptively highlight important features and spatial attention mechanisms to focus on crucial regions in the feature map. This design ensures that GADRM effectively integrates multi-level feature enhancement, improving its capacity for discriminating complex IoT attack patterns.

Stage I: The input feature map M is first partitioned into two subgroups along the channel dimension, where subgroup M_1 retains $\frac{3}{4}$ of the channels and subgroup M_2 contains the remaining $\frac{1}{4}$. Subsequently, feature map M undergoes a 1×1 convolution operation followed by channel-wise grouping, generating eight distinct feature map groups denoted as $Group_i$ (with each group $i = 1, 2, \dots, 8$ containing $\frac{1}{8}$ of the input feature map channels).

Stage II: We divide each feature map $Group_i$ into 4 groups along the channel dimension (each group of feature maps is represented as X_r , where $r = 1, 2, 3, 4$). Firstly, the input feature map X_r undergoes channel shuffling, followed by a Reshape operation to transform the input channels from one dimension into two dimensions. One dimension represents the convolution groups, and the other dimension corresponds to the channels in each convolution group. Next, operation Transpose is performed to swap the two expanded dimensions. Finally, operation Flatten is applied to flatten the result, completing the channel shuffling process.

We use $DWConv(\cdot)$ to represent the 3×3 depthwise separable convolution in GADRM, and Y_r denotes the output of $DWConv(\cdot)$. The mutual dependencies between channel mappings can enhance the feature representation of specific semantics. The feature map Y_r from the r th group is input into the SE module. First, the spatial dimensions of the feature layer are compressed using adaptive global average pooling (GAP). Then, two fully connected layers ($FC_{Sigmoid}$ and FC_{Relu}) are employed to weigh the channel information, resulting in the weights for each group of channels:

$$W_r = FC_{Sigmoid}(FC_{Relu}(GAP(Y_r))) \quad (5)$$

Finally, the weights of each channel are multiplied with each grouped feature map to obtain a channel-weighted feature map:

$$G_i = W_r \times Y_r \quad (6)$$

In Eqs. (5)–(6), $r = 1, 2, 3, 4$ and $i = 1, 2, \dots, 8$.

Stage III: The spatial attention module is employed to aggregate spatial relationships, enabling the model to focus more on the key features of IoT attacks. First, the feature maps G_i obtained from all groups are summed to obtain the overall feature map:

$$G = \sum_{i=1}^8 G_i \quad (7)$$

Next, global average pooling (GAP) and global max pooling (GMP) are applied to the feature map G to obtain two different types of contextual information. The two feature maps, $GAP(G) \in \mathbb{R}^{1 \times H \times W}$ and $GMP(G) \in \mathbb{R}^{1 \times H \times W}$, are then concatenated and passed through a 3×3 convolution operation to obtain the weights for the spatial dimensions of the feature map G :

$$W = Conv(Concat[GAP(G), GMP(G)]) \quad (8)$$

In Eq. (8), $Concat$ represents the feature map concatenation operation, and $Conv$ denotes a standard 3×3 convolution operation. Finally, an element-wise multiplication is performed between the weight map W and the input feature map G to obtain the final spatial attention map.

Stage IV: Finally, the feature map N undergoes a 1×1 convolution layer operation and is then divided into two sections based on the channels. Three-quarters of the channels of the feature map N_1 are added to the lower skip connection path and are summed with the initial feature map M_1 . The remaining one-quarter of the channels of the feature map are concatenated with the initial module of the upper skip connection path. The channels are then merged to form the output:

$$Output = Concat[Concat[M_2, N_2], M_1 + N_1] \quad (9)$$

In Eq. (9), $Output$ represents the final output of the module, and $Concat$ denotes the concatenation operation of the feature maps along the channel dimension.

3.1.5. GLFEM strategy construction

In the context of IoT intrusion detection, the term spatial relationships do not refer to physical space as seen in natural images, but rather to the structured arrangement and contextual dependencies among feature dimensions. IoT network traffic data typically consists of multi-dimensional statistical features, which exhibit inherent semantic correlations such as protocol behaviors and traffic patterns, forming structured relationships across different dimensions. These relationships, although not spatial in the conventional sense, can be modeled similarly to spatial neighborhoods in feature maps. Therefore, we design a GLFEM in the deeper layers of the network to simultaneously extract local and global information. This module performs global modeling of high-level semantic information for IoT attacks, enabling the model to effectively extract key features, thereby improving the network performance for unknown attack categories. The GLFEM module consists of three sub-modules: the local information encoding module, the global information encoding module, and the feature fusion module (see Fig. 4).

To model these relationships, GLFEM applies a block partitioning strategy, where the feature map is divided into smaller blocks. Self-attention is then applied within each block to capture local dependencies. This process enables the model to learn feature correlations within each block while maintaining computational efficiency. Additionally, the global dependencies are captured through the Transformer, which encodes relationships across blocks, allowing for the modeling of global semantic features. This approach forms a local-global relational network, where both local and global dependencies are integrated into a unified feature representation, enabling the model to capture complex attack patterns in IoT data.

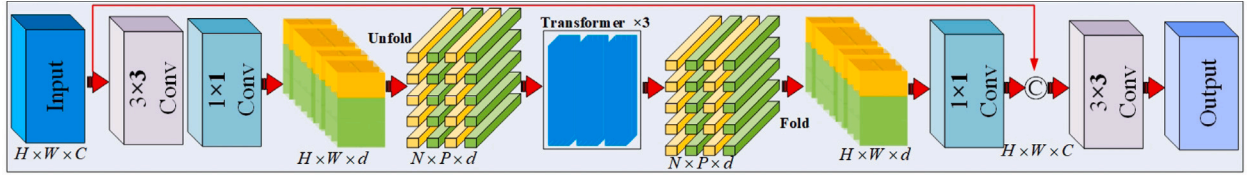


Fig. 4. GLFEM module.

First, the feature map $X \in \mathbb{R}^{H \times W \times C}$ enters the local information encoding module, where a convolutional layer with a kernel size of 3×3 is used for local information encoding. Then, a 1×1 convolutional layer is applied to adjust the channels. Additionally, the tensor is projected into a high-dimensional space by learning a linear combination of the input channels, denoted as $X \in \mathbb{R}^{H \times W \times d}$.

To enable the network to learn global representations with spatial inductive bias, the input feature map is first divided into patch blocks, with the feature map represented as X_U , where $X_U \in \mathbb{R}^{P \times N \times d}$ and $P = WH$ holds. Here, N is the number of blocks, $N = HW/P$, with H and W representing the height and width of each block, respectively. Specifically, the feature maps with the same relative positions are concatenated into a block through the operation *Unfold*, and self-attention computation is performed within each block to reduce the computational complexity. Meanwhile, we input the concatenated sequence into a Transformer for modeling.

Finally, the Transformer is used to encode the inter-relationships, and the representation is computed as:

$$X_G(p) = \text{Transformer}(X_U(p)), \quad 1 \leq p \leq P \quad (10)$$

After self-attention computation, the feature map group is restored to its original shape via the *Fold* operation. This module also retains the order of patches. Simultaneously, the globally modeled feature blocks X_G are passed through a 1×1 convolutional layer to adjust the channel count back to its original size. A residual connection is then applied, concatenating the result with the original input feature map along the channel dimension. The final output is obtained through feature fusion via a 3×3 convolutional layer.

In conclusion, the proposed GADRM utilizes a dual-residual structure with grouped attention to enhance feature interaction and preserve multi-path information flow. This design improves semantic focus and enables more discriminative IoT attack representations while minimizing redundancy. On the other hand, GLFEM combines local convolutional encoding and global attention modeling to integrate fine-grained details and high-level semantics, allowing for effective learning even with limited data. Together, these modules complement each other by extracting local and global features, forming a unified feature representation that significantly enhances the generalization and adaptability to previously unseen IoT attacks.

3.2. Adaptive incremental classifier

Since old-class data is unavailable during incremental learning, it is necessary to establish global dependencies between the prototypes and feature representations of old and new tasks, to capture cross-task differences and support adaptive adjustment in the shared feature space. To maintain the stability of the discriminative structure learned during pretraining, FAN is frozen at this stage, which helps prevent catastrophic forgetting caused by direct backbone updates. On this basis, we design an adaptive incremental classifier (AIC) within a meta-training framework, consisting of the hybrid relational projection network (HRP) and the pseudo-episode selection (PES) scheme. In this way, FAN provides a stable representation foundation, while HRP and PES together endow the model with transferable adaptation capability for subsequent incremental recognition tasks.

Algorithm 1 Dual Res-Transformer

Require: D_{base} : Base dataset containing labeled samples for all known categories

Require: D_{new} : New category dataset containing labeled samples

Ensure: P_{base} : Prototypes for the known categories

1: FAN Pretraining Phase:

2: Initialize FAN model (Dual Res-Transformer with GADRM and GLFEM modules)

3: **for** each epoch **do**

4: **for** each batch in D_{base} **do**

5: Apply channel partition, shuffling, and depth-wise separable convolution according to Eq. (5)

6: Apply squeeze-and-excitation and spatial attention according to Eq. (6)

7: Perform local and global features encoding according to Eq. (10)

8: Fuse local and global features $X_G(p)$

9: **end for**

10: **end for**

11: Compute category prototypes for each category W_{base}

3.2.1. Hybrid relational projection network (HRP)

First, we present a hybrid relational projection network (HRP), which integrates the prototype self-projection network (PSP) and the feature representation cross-mapping network (FCP), as shown in Fig. 5. To maintain prototype consistency during incremental learning, we introduce a transformer-based relational projection mechanism that models dependencies between old and new class prototypes. Unlike direct prototype replacement or independent updates, this strategy performs semantically guided reconstruction in a shared relational space, which improves adaptation while reducing the risk of representation drift.

Considering that the goal of PSP is to adaptively adjust prototypes to improve their distinguishability within the current task space, we design the PSP based on Transformer to train a projection function $T(\cdot)$, which transforms a set of original prototypes W into a set of updated prototypes W_{new} (where $W_{new} = T(W)$ holds). In the incremental learning phase, the prototype update strategy is not based on random transformations. Instead, it utilizes the Transformer attention mechanism to establish relationships between old and new class prototypes, ensuring a semantically guided update. This process avoids the instability typically associated with frequent transformations by using a relationship-aware structured adjustment rather than introducing random drift. Furthermore, the hybrid mechanism of PSP and FCP maintains consistency between prototypes and feature representations through a bidirectional projection, which ensures stability and minimizes representation drift. The input for $T(\cdot)$ is in the form of a triplet consisting of the query vector Q , key vector K , and value vector V , denoted as (Q, K, V) . Additionally, Q, K, V is obtained through a linear transformation of the input source W . The PSP can be expressed as:

$$Q = W_Q W, \quad K = W_K W, \quad V = W_V W \quad (11)$$

$$\tilde{V} = \text{softmax}\left(\frac{QK^T}{\sqrt{m}}\right)V \quad (12)$$

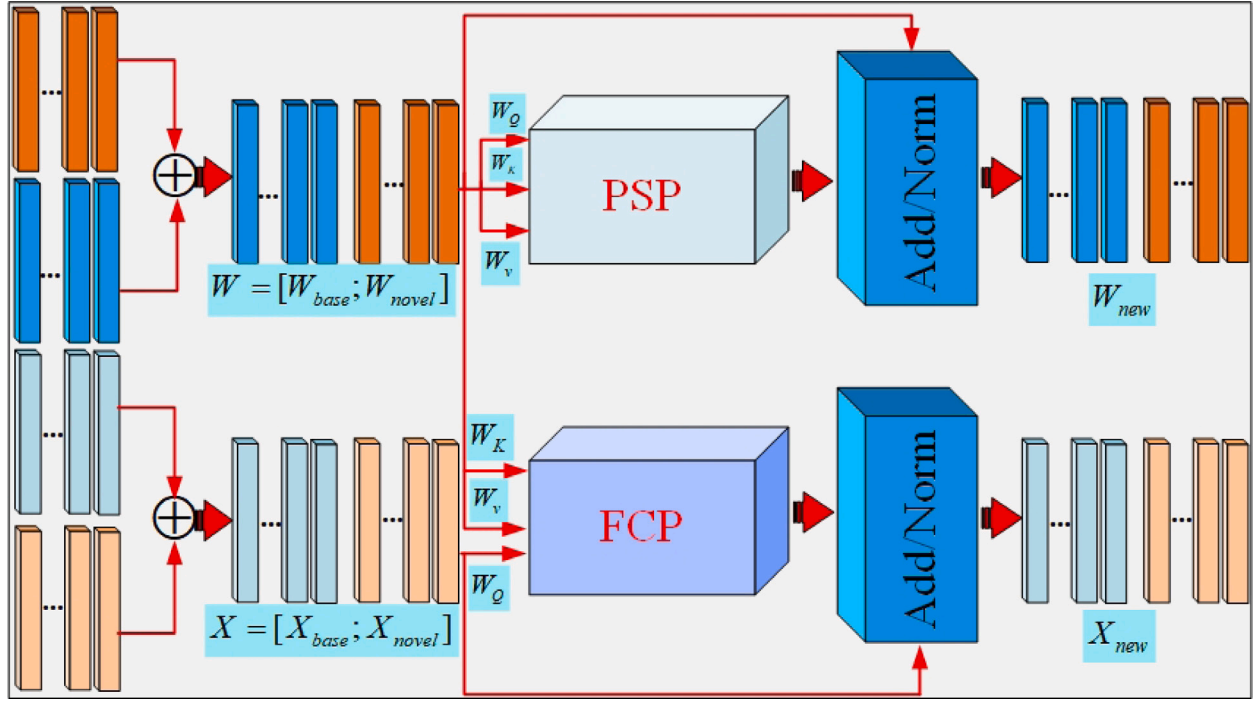


Fig. 5. HRP module.

$$W_{new} = \text{LayerNorm}(\text{Dropout}(W_{FC}\tilde{V} + W)) \quad (13)$$

In Eq. (13), $W = [W_{base}; W_{novel}]$ represents the merging of old and new category prototypes, which are simultaneously updated through PSP. $W_Q, W_K, W_V \in \mathbb{R}^{hm \times m}$ is the learnable transformation matrix that maps the original prototypes to a shared metric space. $W_{FC} \in \mathbb{R}^{m \times hm}$ is the trainable weight matrix, where m denotes the dimensionality of the feature vectors. Specifically, the relevance between each prototype is computed through the operation $\text{softmax}(QK^T/\sqrt{m})$ to obtain the relationship matrix. We then use the relationship matrix as attention weights to weight all prototype information in V , resulting in a discriminative representation $\tilde{V}$. Finally, the updated prototypes W_{new} are obtained by merging with the original prototypes.

Due to the inevitable mismatch between the newly updated prototype representations and the features of both old and new categories, as well as the classifier weights, it is necessary to establish a cross-mapping between the sample features X and prototypes W to match the sample features to their corresponding prototypes. FCP and PSP use the same computational form (i.e., $X_{new} = T(X, W)$ holds), and the input for $T(\cdot)$ still follows the triplet format. Q is obtained through a linear transformation of X , and K, V is obtained through a linear transformation of W . The FCP can be expressed as:

$$Q = W_Q W, \quad K = W_K W, \quad V = W_V W \quad (14)$$

$$\tilde{X} = \text{softmax}\left(\frac{QK^T}{\sqrt{m}}\right)V \quad (15)$$

$$X_{new} = \text{LayerNorm}(\text{Dropout}(W_{FC}\tilde{X} + X)) \quad (16)$$

In Eq. (16), $X = [X_{base}; X_{novel}]$ represents the merging of old and new category prototypes, and the correlation between the prototypes is computed through the operation $\text{softmax}(QK^T/\sqrt{m})$. Then, the information of all prototypes in V is weighted using attention weight coefficients. The weighted information is merged with the original representation to obtain the updated X_{new} .

3.2.2. Pseudo-episode selection (PES)

In practical FSCIL settings, real unseen attack categories are unavailable during training, making it difficult to simulate the distribution

shifts encountered during deployment. Meanwhile, effective HRP training requires continual adaptation to evolving class distributions. To address this, we introduce a pseudo-episode selection (PES) strategy to generate controllable pseudo-incremental tasks for multi-stage meta-training. Unlike traditional Mixup, which is used for data-level regularization within fixed tasks, PES creates semantically meaningful proxy categories within the feature space. PES interpolates between base category data to simulate future unseen categories, enabling HRP to acquire transferable adaptation capabilities before encountering real incremental categories. These pseudo-incremental tasks not only smooth the training distribution but also mimic potential category transitions and support task-level adaptation in HRP. PES extends the Mixup principle to the task level, facilitating continual learning under data scarcity and open-set environments. PES randomly selects two samples, I_a and I_b from two different base categories (a and b) to generate a representative pseudo-incremental category (PIC) sample I_{ab} :

$$I_{ab} = \lambda I_a + (1 - \lambda) I_b \quad (17)$$

In Eq. (17), λ is a random interpolation coefficient, and using N_b base categories, $N_b(N_b - 1)/2$ pseudo-incremental categories can be generated, where $\lambda \sim \text{Beta}(\alpha, \alpha)$ and $\alpha = 0.4$. Similar to the episodic learning paradigm, a pseudo-incremental support set (PIS) is constructed in an N_n -way K -shot manner from the pseudo-incremental dataset, and the corresponding pseudo-incremental query set (PIQ) is sampled from the same N_n categories with E samples each. Enforcing $a \neq b$ ensures that the generated pseudo-sample I_{ab} lies within the convex hull between distinct category manifolds, thus forming novel samples that emulate unseen incremental categories rather than simple within-category perturbations.

Under a local linearity assumption of the embedding function $\phi(\cdot)$:

$$\phi(I_{ab}) \approx \lambda \phi(I_a) + (1 - \lambda) \phi(I_b) \quad (18)$$

It indicates that PES expands the inter-class support region in the embedding space and enhances feature diversity. Furthermore, assuming the loss function $L(\cdot)$ is approximately convex with respect to the model outputs, we obtain:

$$L(g(\phi(I_{ab}))) \leq \lambda L(g(\phi(I_a))) + (1 - \lambda) L(g(\phi(I_b))) \quad (19)$$

By constructing pseudo-incremental tasks, PES regularizes the training process under a vicinal risk minimization framework and exposes the model to smoother inter-class transitions during meta-training. This mechanism enables HRP to learn a more stable relational adaptation pattern, thereby improving its generalization and robustness in subsequent real incremental tasks. To avoid generating unrealistic interpolated samples, we further adopt time-channel decomposition and segment-level mixing for sequence-level synthesis. Instead of applying global linear interpolation over the entire sequence, interpolation is performed within localized temporal regions, which preserves the temporal structure and dynamic characteristics of network traffic. As a result, the generated pseudo-samples are more structurally plausible and semantically consistent, allowing PES to provide meaningful proxy episodes for adapting to unseen categories.

Algorithm 2 FAN-AFSIL model

Require: D_{base} : Base dataset containing labeled samples for all known categories

Require: D_{new} : New category dataset containing labeled samples

Require: P_{base} : Prototypes for the known categories

Ensure: Classification results for the test set, including both known and open-set (unknown) categories

```

1: Incremental Learning stage:
2: for  $t \in \{1, \dots, T\}$  do
3:   for each batch in  $D_{base}$  do
4:     Apply prototype self-projection (PSP) to adjust prototypes
       according to Eqs. (11)–(13)
5:   Update prototypes by merging old and new prototypes
6:   Perform feature representation cross-projection according to
       Eqs. (14)–(16)
7:   Compute loss based on updated prototypes and feature
       representations
8:   end for
9: end for
10: Generate pseudo-incremental category (PIC) by interpolating
     between samples from  $D_{base}$  according to Eq. (17)
11: Return updated AIC model after incremental learning phase
12: Open-set recognition stage:
13: for each sample in  $D_{test}$  do
14:   Extract sample representation using FAN model (FAN features
       are frozen) according to Eq. (5)
15:   if new attack category detected then
16:     Feed the new attack prototype and base prototypes into PSP
       module for update according to Eq. (13)
17:   end if
18:   Obtain updated representation by inputting the attack repre-
       sentation, its prototype, and base prototypes into the PES module
       according to Eqs. (14)–(16)
19:   Classify the sample using the updated representation through
       NCM classifier according to Eq. (3)
20:   Merge the updated representation into base prototype for future
       tasks according to Eq. (4)
21: end for
22: Return final classification results (known or unknown)

```

4. Experiments and result analysis

4.1. Datasets and evaluation metrics

4.1.1. Datasets

To comprehensively evaluate the model's performance, we conducted experiments on two representative IoT datasets, Gas Pipeline and Water Storage Tank (Wu et al., 2024), along with two widely used network datasets, CICIDS2017 and NSL-KDD (Zeng et al., 2025). Gas Pipeline and Water Storage Tank are widely used in industrial

IoT, particularly for monitoring MODBUS protocol traffic. Each dataset contains eight categories: one normal category and seven attack categories (NMRI, CMRI, MSCI, MPCI, MFCI, DoS, and Recon). NMRI and CMRI are response injection attacks that interfere with system behavior by simulating legitimate responses. MSCI, MPCI, and MFCI represent command injection attacks designed to alter control commands. DoS attacks disrupt communication between the control system and the process, while Recon involves reconnaissance attacks aimed at gathering system information. CICIDS2017 is a real-world network traffic dataset designed for evaluating intrusion detection systems. It spans five days of traffic data, containing both normal traffic and six types of attacks: Bot, BruteForce, DoS, Infiltration, PortScan, and WebAttack. Bot refers to malicious traffic generated by botnets, BruteForce focuses on password cracking via brute-force attempts, DoS consumes system resources, Infiltration denotes unauthorized internal access, PortScan is used for port scanning, and WebAttack targets web application vulnerabilities. The NSL-KDD dataset simulates network and intrusion activities. It includes one normal category and four attack types: DoS, Probe, R2L, and U2R. DoS attacks aim to interrupt communication, Probe is used for information gathering, R2L involves remote-to-local access attempts, and U2R consists of attacks that escalate user privileges to system-level access (root).

The numbers of samples in each dataset were as follows: NSL-KDD (original: 148,514; balanced: 29,000), Gas Pipeline (original: 97,019; balanced: 50,205), CICIDS2017 (original: 56,661; balanced: 37,126), and Water Storage Tank (original: 236,179; balanced: 57,000). Each balanced dataset was then randomly split into training, testing, and validation sets with a ratio of 70%:20%:10%. To standardize the evaluation under the few-shot categories-incremental learning setting, we divide the categories in each dataset into base categories and novel categories. As shown in Table 1, the base categories used for training are defined as follows: NSL-KDD, 4 categories (1 normal category and 3 attack categories); Gas Pipeline, 6 categories (1 normal category and 5 attack categories); CICIDS2017, 5 categories (1 normal category and 4 attack categories); and Water Storage Tank, 5 categories (1 normal category and 4 attack categories). The remaining unseen attack categories are treated as novel categories and are introduced only during testing and incremental evaluation. Accordingly, the category configurations for training and testing are: NSL-KDD (training: 4 categories, testing: 5 categories), Gas Pipeline (training: 6 categories, testing: 7 categories), CICIDS2017 (training: 5 categories, testing: 6 categories), and Water Storage Tank (training: 5 categories, testing: 6 categories). This setting ensures that the training set contains only known categories, while the testing stage introduces previously unseen attack categories for evaluating generalization and open-set recognition capability.

Following the FSCIL protocol, the incremental learning process is organized into multiple sessions. Specifically, NSL-KDD contains 1 incremental session, Gas Pipeline contains 1 incremental session, CICIDS2017 contains 1 incremental session, and Water Storage Tank contains 1 incremental session, where each session introduces one novel attack category with only a few labeled samples. The model is evaluated after each session to assess its ability to incorporate new categories while preserving discrimination for previously learned categories and unknown attacks. For the few-shot setting, we adopt two configurations with $K \in \{10, 20\}$. In each session, 10 or 20 labeled samples are randomly selected from each novel category to form the support set, simulating real-world security scenarios where only a few labeled samples are available for emerging attacks, while the remaining samples are used to form the query/test set. The support set is used for incremental adaptation, and the query set is used for performance evaluation after each session. Although the attack categories treated as novel or unknown differ across datasets due to their different class compositions, all datasets follow the same FSCIL-based protocol, few-shot setting, and evaluation procedure. To further ensure statistical reliability, each configuration was repeated 20 times with different random splits.

Table 1
Division of known and unknown attack types in each dataset.

Dataset	Attacks	Known	Unknown
NSL-KDD	DoS, Probe, R2L, U2R	DoS, Probe, U2R	R2L
Gas Pipeline	NMRI, CMRI, MSCI, MPCl, MFCl, DoS, Recon	CMRI, MSCI, MPCl MFCl, Recon	NMRI, DoS
CICIDS2017	Bot, BruteForce, DoS, Infiltration, PortScan, WebAttack	Bot, BruteForce, DoS, Infiltration	PortScan, WebAttack
Water Storage Tank	NMRI, CMRI, MSCI, MPCl, MFCl, DoS, Recon	MSCl, MPCl, MFCl, DoS	NMRI, CMRI, Recon

4.1.2. Evaluation metrics

To evaluate the effectiveness of the FAN-AFSIL framework, we utilize four evaluation metrics, as detailed below:

(1) **AUROC** represents the area under the ROC curve. The ROC curve illustrates the relationship between the true positive rate ($TPR = TP / (TP + FN)$) and the false positive rate ($FPR = FP / (FP + TN)$).

(2) **AUPRI** represents the area under the precision–recall curve for the known data designated as the positive category. The precision–recall curve is a graphical representation that illustrates the relationship between precision ($TP / (TP + FP)$) and recall ($TP / (TP + FN)$).

(3) **AUPROUT** denotes the area under the precision–recall curve for the unknown data designated as the positive category.

The detection error is used to measure the mis-classification probability when TPR is set at 95%, calculated as $P_e = 0.5(1 - TPR + 0.5FPR)$.

In the experiments, all known categories detected by the FAN-AFSIL, except for AUPROUT, are considered as the positive categories, while the detected unknown categories are regarded as the negative category. In practice, these four metrics are utilized to evaluate performance, with the first three metrics approaching 1 and the P_e approaching 0 indicating better performance.

To further assess the effectiveness of the FAN-AFSIL in detecting unknown intrusions compared to known intrusions, we also select accuracy as an additional evaluation indicator. Specifically, we use Acc_k to represent the accuracy for known intrusion categories and Acc_u to denote the accuracy for unknown intrusion categories. Moreover, we also use $H_{score} = \frac{2 * Acc_k * Acc_u}{Acc_k + Acc_u}$ to synthesize a measure of performance. Finally, openness (Scheirer et al., 2013) is a key factor in constructing the test dataset, as it reflects the proportion of unknown attacks to known categories in the test data. A high degree of openness means limited understanding of known attack behaviors during training, which makes it more challenging to distinguish between known and unknown attacks. To further investigate the impact of openness on performance, experiments were conducted under different openness settings. The degree of openness is defined as:

$$Openness = 1 - \sqrt{\frac{2 * N_{train}}{N_{test} + N_{target}}} \quad (20)$$

In Eq. (20), N_{train} represents the categories observed during training, N_{test} denotes the categories to be observed during testing, and N_{target} refers to the number of target categories that need to be correctly identified during the testing process.

Openness is a parameter used in classification problems. A value of 0 indicates a fully known closed-set classification, while higher values suggest an increased likelihood of encountering unknown attacks.

To evaluate the model's robustness to noise, Gaussian noise was added to the original signals to create composite signals with varying signal-to-noise ratios (SNRs). This approach allowed for the assessment of the impact of different noise environments on model performance. The SNR was calculated as shown in Eq. (21), where P_{signal} represents the power of the valid signal and P_{noise} represents the power of the noise signal.

$$SNR = 10 \times P_{signal} / P_{noise} \quad (21)$$

4.2. Model parameterization and implementation details

In the pre-training phase of FAN, ResNet-34 is used as the backbone architecture for feature extraction on Datasets I–II. The initial learning rate is set to 0.01, with adjustments made at the 30th and 60th iterations, reducing it to 0.001 and 0.0001, respectively. The SGD optimizer is employed for 100 iterations, with a batch size of 32. For Datasets III and IV, ResNet-18 is used as the backbone network, with an initial learning rate set to 0.1, which is adjusted to 0.01 and 0.001 after the 30th and 40th iterations, respectively. ResNet-18 and ResNet-34 were selected to represent lightweight and moderate deployment scenarios, reflecting FAN's performance under different resource budgets. ResNet-18, with its compact structure, is suitable for low-latency inference on resource-constrained devices, while ResNet-34 offers better representation capability at a moderate computational cost, making it ideal for edge servers requiring higher accuracy. Additionally, the modular GADRM and GLFEM modules in FAN, which interface with feature maps, can be easily integrated with various CNN backbones, allowing seamless adaptation to both lighter and more powerful architectures, enhancing deployment flexibility. Experimental results show that an initial learning rate of 0.01 yielded the best performance across different datasets. As a result, we further investigate the effects of different learning rate configurations in subsequent experiments. In the adaptive incremental learning phase, the support set for Datasets I–II is configured as 5-way, while for Datasets III–IV, the support set is configured as 10-way. The query set size for all four datasets is fixed at 10. The initial learning rate is uniformly set to 0.0002, and the HRP module is trained for 100 iterations. These parameters were found to be sensitive to the specific characteristics of the datasets and tasks. A learning rate of 0.01 was optimal for most datasets, while the query set size also had a significant impact on performance, depending on the task at hand. The results of this tuning process will be further elaborated in subsequent experiments, where the effects of different parameter configurations will be explored to ensure the optimal settings for each scenario.

4.3. Open-set IoT attack detection

4.3.1. IoT known attack detection experiments

To evaluate the performance of FAN-AFSIL in known intrusion detection, we conduct confusion matrix analyses across four datasets under two few-shot settings (10-shot and 20-shot). Based on these results, we further introduce unknown intrusion categories into each dataset to simulate a realistic open-set environment, thereby evaluating the model's ability to accurately recognize known attacks in the presence of unseen ones. As shown in Fig. 6, FAN-AFSIL consistently achieves strong detection performance for known intrusion categories across different datasets, maintaining high accuracy even under limited samples. After incorporating IoT-related unknown categories, it continues to exhibit stable detection performance with only minimal degradation, demonstrating its robustness and generalization ability in open-set intrusion detection scenarios.

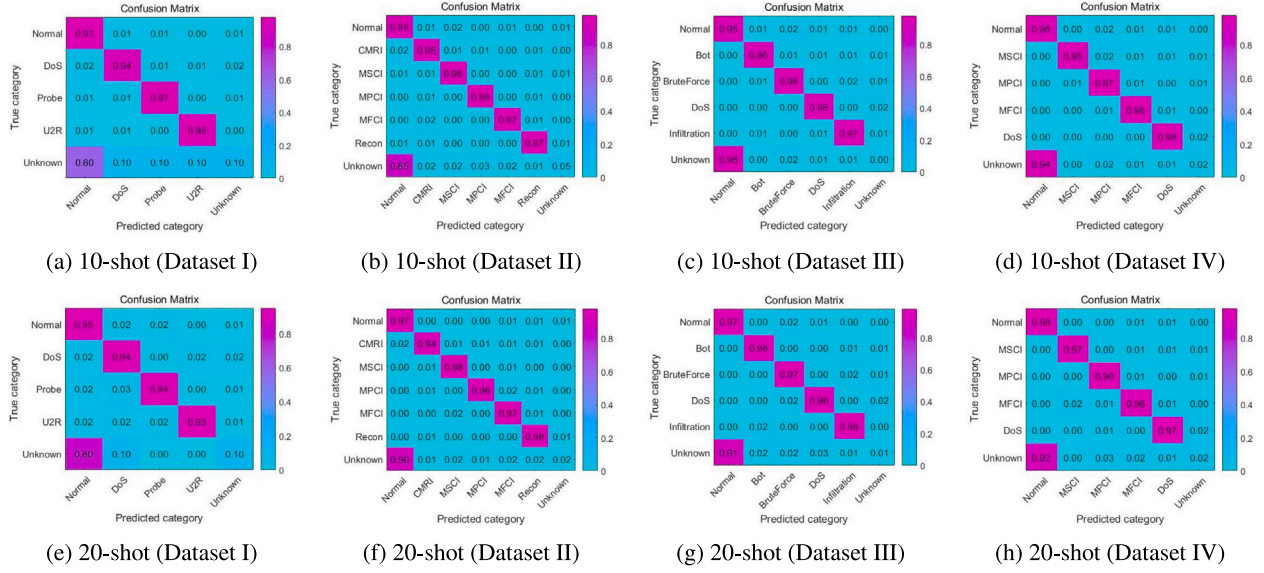


Fig. 6. Results of FAN-AFSIL of IoT known attack detection.

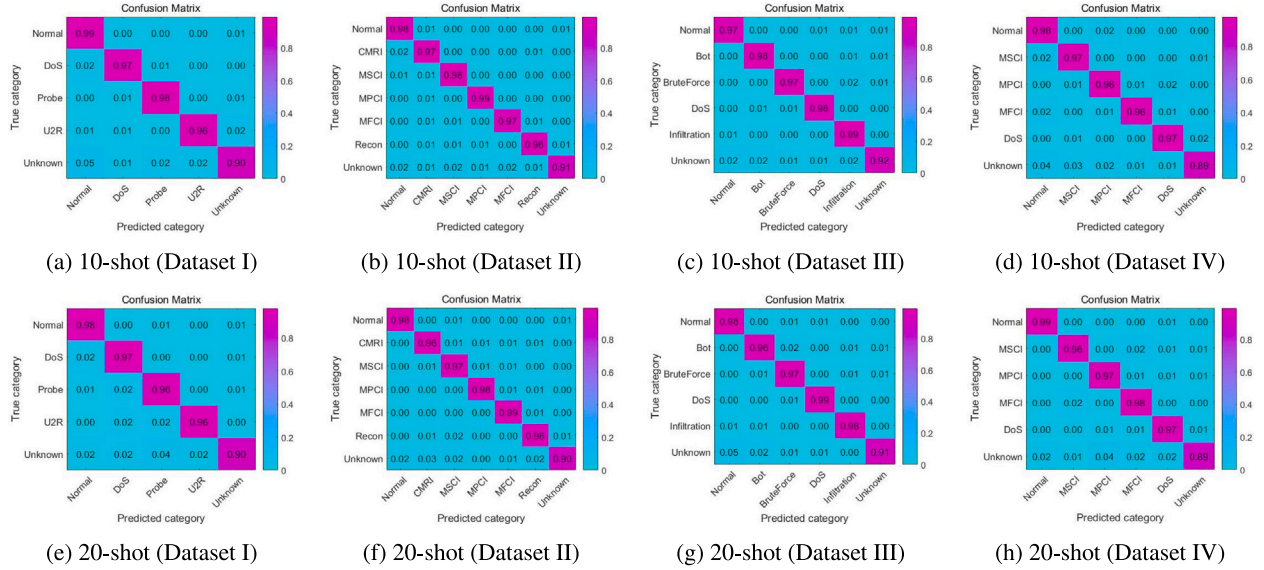


Fig. 7. Results of FAN-AFSIL detection of a single unknown intrusion.

4.3.2. IoT unknown attack detection experiments

To further evaluate the performance of FAN-AFSIL in unknown intrusion detection, we conduct confusion matrix experiments across four datasets, testing under two few-shot settings (10-shot and 20-shot). Building upon these experiments, the FAN-AFSIL model's detection performance is optimized not only for known attack types but also validated for its performance in the presence of unknown attack types. As shown in the experimental results presented in Fig. 7, FAN-AFSIL consistently maintained a detection accuracy of over 0.95 for unknown attacks across all four datasets, indicating its strong performance in detecting unknown attack types. Regardless of whether the setting is 10-shot or 20-shot, FAN-AFSIL can sustain a high detection capability for unknown attacks. These results demonstrate that FAN-AFSIL excels not only in known intrusion detection but also remains highly effective in detecting IoT unknown attacks in an open-set environment, showcasing its robustness and adaptability.

4.3.3. Comparison of detection performance with the baseline model

To thoroughly evaluate the performance of the FAN-AFSIL model, we conduct ROC curve analysis using three baseline models. In this evaluation, we perform classification tasks on Datasets I–IV under a 20-shot setting, focusing on distinguishing between normal and unknown categories. The selected baseline models include FSIL (a basic model with no optimization strategies applied), prototype compositional networks (PCN), and EVM (an open-set recognition model based on extreme value theory). Specifically, all comparative models used the same backbone network (ResNet-34). The training protocol was consistent across all models: the initial learning rate was set to 0.01, with adjustments made at the 30th and 60th iterations, reducing it to 0.001 and 0.0001, respectively. The SGD optimizer was employed for a total of 100 iterations with a batch size of 32. Importantly, the training set only contained known attack categories, while the test set introduced attack types that had not been encountered during training, enabling the evaluation of the models' generalization and

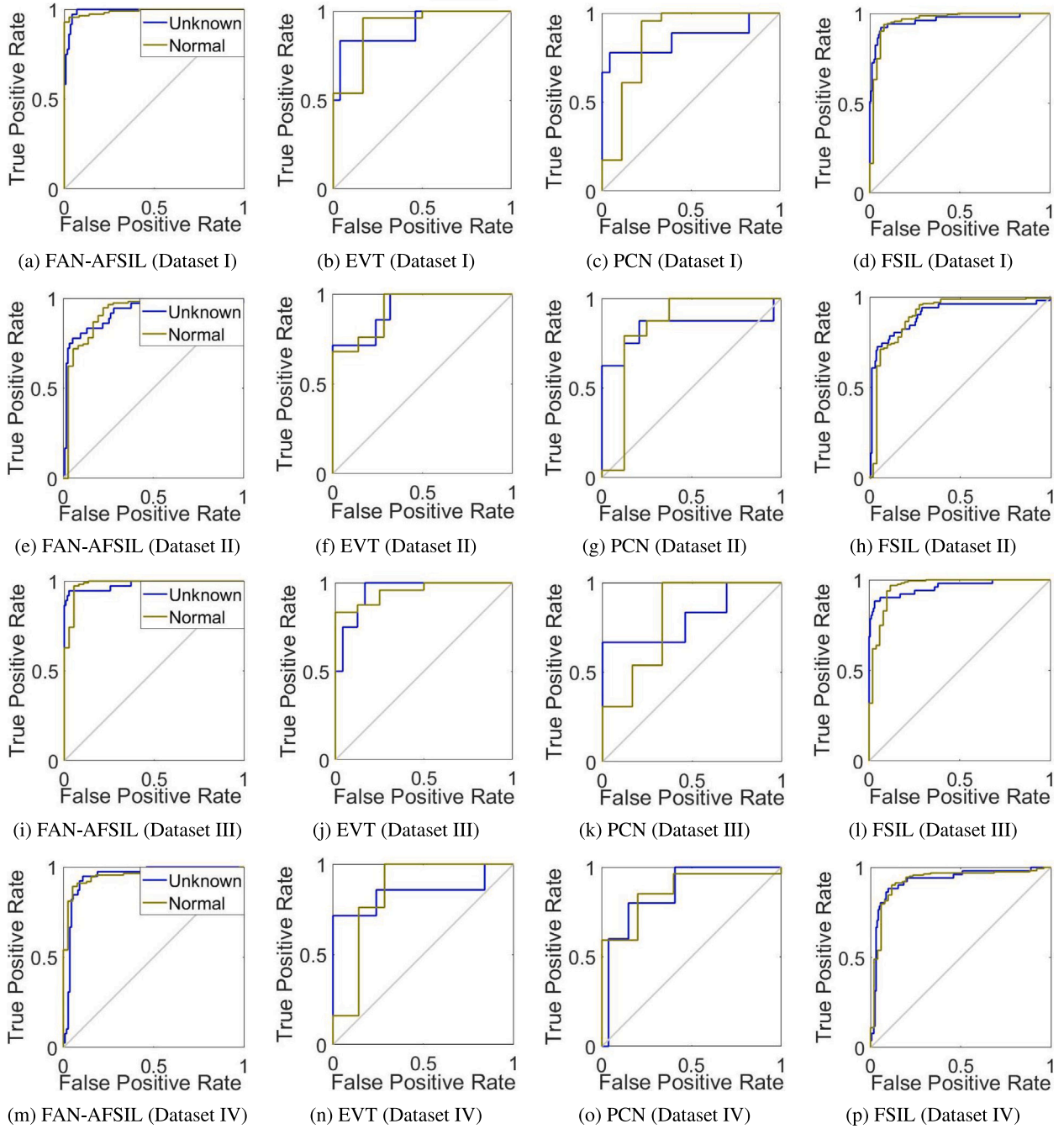


Fig. 8. ROC curve comparison of the proposed models across four datasets.

open-set recognition capabilities. The detailed experimental results for each model are illustrated in Fig. 8. As shown, FAN-AFSIL consistently outperforms the baseline models (FSIL, PCN, and EVT) across four test datasets (Datasets I–IV) in detecting unknown-category intrusions in IoT environments. Moreover, this performance advantage is not limited to a single dataset but is observed uniformly across different experimental settings. When confronted with unseen attack categories, FAN-AFSIL achieves higher accuracy and exhibits stronger robustness.

To evaluate the overall performance of the FAN-AFSIL model, we conduct a systematic comparison with three baseline models in terms of metrics Acc_k , Acc_u , and H_{score} . To assess the practical applicability of the models, we also report the runtime and memory (which measures the memory usage of the model during operation or inference, with smaller values indicating better performance) during the detection

process. Specifically, we report on the detection metrics of the four models across four datasets, as shown in Table 2.

The experimental results demonstrate that FAN-AFSIL achieved the best performance on all datasets, particularly in metrics Acc_k and Acc_u , where its values consistently exceeded 0.96 and 0.93, significantly outperforming the other baseline models. This indicates that FAN-AFSIL not only achieves high accuracy in detecting IoT known intrusions but also exhibits strong recognition capabilities in handling IoT unknown intrusions.

Although its runtime and memory are slightly higher compared to some baseline models, it remains within an acceptable range for practical applications. Therefore, these results further validate the advantages of FAN-AFSIL in efficiently performing IoT intrusion detection in open-set environments.

Table 2
Comparison of detection metrics (10-shot and 20-shot).

model	Dataset	Metrics (10-shot)					Metrics (20-shot)				
		$Acc_k \uparrow$	$Acc_u \uparrow$	$H_{score} \uparrow$	$t/s \downarrow$	Memory /MB↓	$Acc_k \uparrow$	$Acc_u \uparrow$	$H_{score} \uparrow$	$t/s \downarrow$	Memory /MB↓
PCN	Dataset I	0.8834	0.7612	0.8178	52.14	89.6124	0.8691	0.7962	0.8311	53.69	87.6017
	Dataset II	0.8407	0.7756	0.8068	56.37	90.8547	0.8554	0.7886	0.8206	55.17	88.2385
	Dataset III	0.8571	0.8063	0.8309	55.19	78.9612	0.8896	0.8158	0.8511	57.20	76.9024
	Dataset IV	0.8412	0.8159	0.8284	59.67	91.2510	0.8695	0.8264	0.8474	58.96	89.5621
EVM	Dataset I	0.8791	0.8361	0.8571	48.91	74.1204	0.8692	0.8355	0.8520	47.63	72.6309
	Dataset II	0.8814	0.8459	0.8633	46.38	75.6368	0.8785	0.8366	0.8570	45.69	76.0058
	Dataset III	0.8936	0.8397	0.8658	47.55	68.1209	0.8861	0.8418	0.8634	48.11	69.3015
	Dataset IV	0.8576	0.8442	0.8508	50.19	65.7526	0.8930	0.8267	0.8586	51.54	70.2557
FSIL	Dataset I	0.8963	0.8226	0.8579	39.17	56.9047	0.9057	0.8311	0.8668	38.66	58.6120
	Dataset II	0.9012	0.8457	0.8726	38.22	60.2338	0.9126	0.8229	0.8654	39.78	62.3078
	Dataset III	0.9145	0.8368	0.8739	40.57	62.8591	0.9230	0.8395	0.8793	41.25	60.3027
	Dataset IV	0.9117	0.8445	0.8768	42.74	59.6785	0.9114	0.8520	0.8807	41.97	58.0165
our	Dataset I	0.9678	0.9217	0.9442	58.63	58.9630	0.9751	0.9364	0.9553	56.47	60.1157
	Dataset II	0.9719	0.9358	0.9535	60.15	62.2054	0.9835	0.9420	0.9623	58.37	64.8963
	Dataset III	0.9636	0.9330	0.9481	59.66	64.1219	0.9762	0.9499	0.9629	57.15	63.4126
	Dataset IV	0.9574	0.9261	0.9415	61.27	60.3788	0.9694	0.9562	0.9628	60.26	60.8524

Table 3
Performance of each SOTA model in detecting different unknown intrusion categories (10-shot).

Model	AUROC $\uparrow$				AUPRIN $\uparrow$				AUPROUT $\uparrow$				$P_e \downarrow$			
	R2L	DoS	Port Scan	NM RI	R2L	DoS	Port Scan	NM RI	R2L	DoS	Port Scan	NM RI	R2L	DoS	Port Scan	NM RI
TM-GAN	0.85	0.86	0.84	0.87	0.82	0.84	0.85	0.84	0.84	0.85	0.81	0.85	0.56	0.41	0.37	0.56
PFTD	0.87	0.88	0.83	0.88	0.85	0.85	0.84	0.85	0.86	0.84	0.82	0.86	0.39	0.33	0.30	0.39
PerFLID	0.86	0.85	0.87	0.89	0.83	0.86	0.88	0.87	0.85	0.83	0.85	0.87	0.42	0.39	0.35	0.42
CTGNN	0.85	0.87	0.85	0.87	0.84	0.88	0.87	0.86	0.87	0.85	0.84	0.86	0.50	0.35	0.31	0.50
VAEEVT	0.90	0.91	0.88	0.92	0.88	0.90	0.90	0.91	0.89	0.90	0.86	0.91	0.33	0.27	0.25	0.33
OSDN	0.89	0.88	0.86	0.90	0.87	0.86	0.88	0.86	0.86	0.85	0.85	0.89	0.38	0.23	0.22	0.38
BLIP	0.83	0.85	0.84	0.85	0.81	0.84	0.86	0.83	0.83	0.84	0.83	0.84	0.31	0.30	0.28	0.31
R-DFCIL	0.91	0.92	0.90	0.92	0.90	0.91	0.91	0.90	0.91	0.89	0.89	0.91	0.40	0.28	0.26	0.40
FACT	0.92	0.93	0.91	0.93	0.91	0.90	0.92	0.91	0.92	0.90	0.90	0.92	0.28	0.21	0.20	0.28
FAN-AFSIL	0.96	0.95	0.94	0.97	0.94	0.93	0.96	0.95	0.96	0.94	0.93	0.96	0.21	0.15	0.11	0.21

4.3.4. Quantitative experimental comparison with state-of-the-art (SOTA) models

To further validate the effectiveness of FAN-AFSIL, we conduct a comparative analysis with current state-of-the-art (SOTA) models based on metrics AUROC, AUPRIN, AUPROUT, and P_e . Specifically, we perform experiments on the unknown intrusion categories defined for each dataset in Table 1 (where Dataset I uses R2L, Dataset II uses DoS, Dataset III uses PortScan, and Dataset IV uses NMRI). For deep learning-based detection methods, we select TM-GAN (Ding et al., 2024), PFTD (X. Yang et al., 2025), PerFLID (Yan et al., 2025), and CTGNN (Duan et al., 2024); For open-set intrusion detection methods, we select VAEVT (Wu et al., 2024) and OSDN (X. Yang et al., 2025). We also implement three incremental learning approaches (BLIP G. Shi et al., 2021, R-DFCIL Gao et al., 2022, and FACT Zhou et al., 2022).

To ensure fairness in the comparative experiments, we standardized the use of ResNet as the backbone network for all models. This approach eliminates potential biases introduced by differing network architectures and training strategies in the original methods. By using a unified backbone, all models are evaluated under consistent conditions, allowing for a more objective comparison of their performance. This design choice ensures that the observed performance differences reflect the models' inherent strengths and weaknesses, rather than external factors. Future work will explore the impact of optimizing each SOTA model with its original configurations to further validate the performance of FAN-AFSIL. Importantly, the training set contained only known attack categories, while the test set introduced attack types that had not been seen during training, enabling the evaluation of the models' generalization and open-set recognition capabilities. To ensure fairness, all baseline models were implemented using custom code and evaluated under the same conditions. Additionally, we employed

a cross-validation-based strategy for unknown intrusion detection. In each round, one intrusion category was designated as the unknown IoT category, while the remaining categories were treated as known categories for training. It ensured that our comparisons were performed in a consistent and reproducible manner, making the results more reliable. The experimental results are presented in Tables 3–4. In this experiment, the FAN-AFSIL model demonstrates the best performance across all unknown categories. For the 10-shot setting, the average values of the FAN-AFSIL model on metrics AUROC, AUPRIN, AUPROUT, and P_e are 0.955, 0.945, 0.947, and 0.17, respectively, outperforming all other SOTA models. This result validates that the FAN-AFSIL can tackle the catastrophic forgetting problem in FSIL without requiring the storage of old category data. Additionally, the model can quickly adapt to new tasks with only a few new samples, achieving superior detection performance for IoT unknown attacks in an open environment. Then, we evaluate the performance of the FAN-AFSIL model in detecting both IoT unknown and known attacks under different incremental tasks, comparing it with current SOTA models. Specifically, Task 1 represents the performance of the baseline model on the base dataset, Task 2 represents the model's update after the arrival of incremental Task 1, where the model is retrained and tested on the current task dataset (which includes both the base dataset and the incremental Task 1), and so on. Therefore, we assess the performance of the above models on different incremental tasks (Task I–IV) for IoT unknown intrusion detection across four datasets. The evaluation metrics include Acc_k , Acc_u , and H_{score} , with the results presented in Tables 6–7. The experimental results show that the FAN-AFSIL model outperforms other models in terms of detection performance for both IoT unknown and known attacks across different incremental tasks. Compared to VAE-EVT and OSDN, which are detection methods in open-set environments, FAN-AFSIL demonstrates significant performance improvements in Acc_k and

Table 4

Performance of each SOTA model in detecting different unknown intrusion categories (20-shot).

Model	AUROC $\uparrow$				AUPRIN $\uparrow$				AUPROUT $\uparrow$				$P_e \downarrow$			
	R2L	DoS	Port Scan	NM RI	R2L	DoS	Port Scan	NM RI	R2L	DoS	Port Scan	NM RI	R2L	DoS	Port Scan	NM RI
TM-GAN	0.86	0.85	0.86	0.88	0.83	0.85	0.86	0.82	0.85	0.86	0.82	0.86	0.52	0.38	0.34	0.52
PFTD	0.85	0.87	0.84	0.89	0.87	0.86	0.85	0.83	0.87	0.85	0.83	0.87	0.41	0.35	0.31	0.36
PerFLID	0.88	0.86	0.88	0.88	0.84	0.87	0.89	0.88	0.88	0.84	0.86	0.88	0.40	0.37	0.33	0.40
CTGNN	0.87	0.88	0.87	0.86	0.85	0.89	0.88	0.89	0.89	0.87	0.85	0.85	0.47	0.33	0.30	0.45
VAEEVT	0.91	0.92	0.89	0.91	0.89	0.91	0.89	0.90	0.88	0.91	0.87	0.90	0.38	0.30	0.23	0.31
OSDN	0.88	0.89	0.87	0.92	0.86	0.88	0.87	0.87	0.87	0.87	0.84	0.90	0.39	0.26	0.21	0.33
BLIP	0.85	0.86	0.83	0.87	0.82	0.87	0.88	0.85	0.84	0.86	0.85	0.85	0.36	0.31	0.26	0.28
R-DFCIL	0.92	0.91	0.91	0.91	0.91	0.92	0.90	0.91	0.90	0.88	0.90	0.90	0.43	0.33	0.24	0.37
FACT	0.93	0.92	0.92	0.92	0.92	0.91	0.91	0.89	0.91	0.91	0.91	0.91	0.30	0.26	0.22	0.29
FAN-AFSIL	0.97	0.96	0.95	0.95	0.96	0.94	0.95	0.94	0.96	0.95	0.94	0.95	0.25	0.17	0.14	0.20

Table 5

Performance of each SOTA model in detecting intrusions on different incremental tasks.

Model	Index	Dataset I				Dataset II			
		Task I	Task II	Task III	Task IV	Task I	Task II	Task III	Task IV
TM-GAN	Acc_k (%) $\uparrow$	89.74 $\pm$ 0.55	88.52 $\pm$ 0.34	86.17 $\pm$ 1.10	83.79 $\pm$ 1.02	89.18 $\pm$ 0.78	88.30 $\pm$ 1.21	87.05 $\pm$ 0.86	84.66 $\pm$ 0.87
	Acc_u (%) $\uparrow$	87.20 $\pm$ 0.11	86.74 $\pm$ 0.09	84.52 $\pm$ 0.11	82.03 $\pm$ 0.83	87.19 $\pm$ 0.26	86.24 $\pm$ 0.50	85.79 $\pm$ 0.37	83.14 $\pm$ 0.05
	H_{score} (%) $\uparrow$	88.45 $\pm$ 0.18	87.62 $\pm$ 0.14	85.34 $\pm$ 0.20	82.90 $\pm$ 0.92	88.17 $\pm$ 0.39	87.26 $\pm$ 0.71	86.42 $\pm$ 0.52	83.89 $\pm$ 0.09
PFTD	Acc_k (%) $\uparrow$	88.79 $\pm$ 0.55	87.16 $\pm$ 0.20	85.72 $\pm$ 0.38	83.80 $\pm$ 0.05	88.60 $\pm$ 1.54	87.41 $\pm$ 1.15	85.89 $\pm$ 1.03	83.11 $\pm$ 1.20
	Acc_u (%) $\uparrow$	87.74 $\pm$ 0.31	86.39 $\pm$ 0.44	84.87 $\pm$ 1.08	82.91 $\pm$ 1.27	87.99 $\pm$ 0.56	86.27 $\pm$ 0.09	84.58 $\pm$ 0.41	82.37 $\pm$ 0.58
	H_{score} (%) $\uparrow$	88.26 $\pm$ 0.40	86.77 $\pm$ 0.28	85.29 $\pm$ 0.56	83.35 $\pm$ 0.10	88.29 $\pm$ 0.82	86.84 $\pm$ 0.17	85.23 $\pm$ 0.59	82.74 $\pm$ 0.78
PerFLID	Acc_k (%) $\uparrow$	90.55 $\pm$ 0.17	89.91 $\pm$ 0.34	87.55 $\pm$ 0.09	85.28 $\pm$ 0.44	90.17 $\pm$ 0.82	89.54 $\pm$ 0.08	87.81 $\pm$ 0.33	84.91 $\pm$ 0.45
	Acc_u (%) $\uparrow$	89.92 $\pm$ 1.06	88.50 $\pm$ 1.26	86.77 $\pm$ 1.11	84.37 $\pm$ 0.35	88.13 $\pm$ 0.63	87.59 $\pm$ 0.77	85.68 $\pm$ 0.54	82.24 $\pm$ 0.09
	H_{score} (%) $\uparrow$	90.23 $\pm$ 0.29	89.20 $\pm$ 0.53	87.16 $\pm$ 0.17	84.82 $\pm$ 0.39	89.14 $\pm$ 0.71	88.55 $\pm$ 0.14	86.73 $\pm$ 0.41	83.55 $\pm$ 0.15
CTGNN	Acc_k (%) $\uparrow$	91.08 $\pm$ 0.33	90.54 $\pm$ 0.87	88.89 $\pm$ 1.27	85.97 $\pm$ 1.01	90.59 $\pm$ 0.80	89.42 $\pm$ 0.05	87.67 $\pm$ 0.30	84.16 $\pm$ 0.09
	Acc_u (%) $\uparrow$	90.72 $\pm$ 0.54	89.91 $\pm$ 0.39	87.58 $\pm$ 0.45	84.52 $\pm$ 0.77	89.51 $\pm$ 0.56	88.37 $\pm$ 1.28	86.55 $\pm$ 0.31	83.74 $\pm$ 0.25
	H_{score} (%) $\uparrow$	90.90 $\pm$ 0.41	90.22 $\pm$ 0.54	88.23 $\pm$ 0.66	85.24 $\pm$ 0.87	90.05 $\pm$ 0.66	88.89 $\pm$ 0.10	87.11 $\pm$ 0.30	83.95 $\pm$ 0.13
VAEEVT	Acc_k (%) $\uparrow$	92.77 $\pm$ 0.96	91.50 $\pm$ 0.35	90.02 $\pm$ 0.42	88.78 $\pm$ 0.50	92.25 $\pm$ 0.33	91.08 $\pm$ 1.47	89.88 $\pm$ 1.15	87.36 $\pm$ 0.44
	Acc_u (%) $\uparrow$	91.59 $\pm$ 1.30	90.88 $\pm$ 0.66	88.94 $\pm$ 0.59	86.67 $\pm$ 1.02	91.14 $\pm$ 0.42	90.25 $\pm$ 0.65	88.92 $\pm$ 0.77	86.89 $\pm$ 0.20
	H_{score} (%) $\uparrow$	92.18 $\pm$ 1.10	91.19 $\pm$ 0.46	89.48 $\pm$ 0.49	87.71 $\pm$ 0.67	91.69 $\pm$ 0.37	90.66 $\pm$ 0.90	89.40 $\pm$ 0.92	87.12 $\pm$ 0.28
OSDN	Acc_k (%) $\uparrow$	92.75 $\pm$ 0.41	91.86 $\pm$ 0.03	90.04 $\pm$ 0.57	88.41 $\pm$ 0.33	93.75 $\pm$ 0.04	92.86 $\pm$ 0.74	91.50 $\pm$ 0.08	89.77 $\pm$ 0.66
	Acc_u (%) $\uparrow$	91.85 $\pm$ 1.02	90.94 $\pm$ 0.85	88.20 $\pm$ 0.15	86.68 $\pm$ 0.05	91.80 $\pm$ 0.62	90.79 $\pm$ 1.26	88.05 $\pm$ 1.13	86.74 $\pm$ 0.81
	H_{score} (%) $\uparrow$	92.30 $\pm$ 0.58	91.40 $\pm$ 0.06	89.11 $\pm$ 0.24	87.54 $\pm$ 0.09	92.76 $\pm$ 0.08	91.81 $\pm$ 0.93	89.74 $\pm$ 0.15	88.23 $\pm$ 0.73
BLIP	Acc_k (%) $\uparrow$	92.89 $\pm$ 1.20	91.75 $\pm$ 0.39	90.04 $\pm$ 0.51	87.50 $\pm$ 0.09	93.16 $\pm$ 0.02	92.48 $\pm$ 0.35	91.07 $\pm$ 0.77	89.24 $\pm$ 0.08
	Acc_u (%) $\uparrow$	91.45 $\pm$ 0.76	90.82 $\pm$ 0.08	88.96 $\pm$ 0.46	85.77 $\pm$ 0.82	92.84 $\pm$ 0.43	91.90 $\pm$ 0.68	90.68 $\pm$ 0.04	88.85 $\pm$ 0.56
	H_{score} (%) $\uparrow$	92.16 $\pm$ 0.93	91.28 $\pm$ 0.13	89.50 $\pm$ 0.48	86.63 $\pm$ 0.16	93.00 $\pm$ 0.04	92.19 $\pm$ 0.46	90.87 $\pm$ 0.08	89.04 $\pm$ 0.14
R-DFCIL	Acc_k (%) $\uparrow$	93.12 $\pm$ 0.59	92.57 $\pm$ 0.09	91.04 $\pm$ 0.50	87.56 $\pm$ 1.27	94.06 $\pm$ 0.78	93.50 $\pm$ 0.04	92.14 $\pm$ 0.53	90.26 $\pm$ 0.45
	Acc_u (%) $\uparrow$	92.79 $\pm$ 1.14	91.80 $\pm$ 1.02	90.51 $\pm$ 0.63	86.96 $\pm$ 0.56	93.44 $\pm$ 0.59	92.61 $\pm$ 0.23	91.09 $\pm$ 0.08	89.87 $\pm$ 0.74
	H_{score} (%) $\uparrow$	92.95 $\pm$ 0.78	92.18 $\pm$ 0.17	90.77 $\pm$ 0.56	87.26 $\pm$ 0.78	93.75 $\pm$ 0.67	93.05 $\pm$ 0.07	91.61 $\pm$ 0.14	90.06 $\pm$ 0.56
FACT	Acc_k (%) $\uparrow$	94.06 $\pm$ 0.56	93.51 $\pm$ 0.08	91.20 $\pm$ 1.36	89.06 $\pm$ 1.02	94.39 $\pm$ 0.50	93.52 $\pm$ 1.34	91.86 $\pm$ 1.01	88.77 $\pm$ 0.35
	Acc_u (%) $\uparrow$	93.31 $\pm$ 0.59	92.09 $\pm$ 0.26	90.85 $\pm$ 0.41	87.93 $\pm$ 0.07	93.37 $\pm$ 0.44	92.75 $\pm$ 0.09	90.93 $\pm$ 0.47	87.86 $\pm$ 0.02
	H_{score} (%) $\uparrow$	93.68 $\pm$ 0.57	92.79 $\pm$ 0.12	91.02 $\pm$ 0.63	88.49 $\pm$ 0.13	93.88 $\pm$ 0.47	93.13 $\pm$ 0.17	91.39 $\pm$ 0.64	88.31 $\pm$ 0.04
FAN-AFSIL	Acc_k (%) $\uparrow$	97.56 $\pm$ 0.03	97.03 $\pm$ 0.86	96.59 $\pm$ 1.40	94.93 $\pm$ 0.77	97.93 $\pm$ 0.41	97.25 $\pm$ 0.56	96.59 $\pm$ 0.78	95.13 $\pm$ 0.09
	Acc_u (%) $\uparrow$	96.74 $\pm$ 0.52	96.14 $\pm$ 0.90	95.48 $\pm$ 0.07	93.75 $\pm$ 0.82	96.50 $\pm$ 0.09	96.03 $\pm$ 0.45	95.63 $\pm$ 0.05	94.82 $\pm$ 0.41
	H_{score} (%) $\uparrow$	97.15 $\pm$ 0.06	96.58 $\pm$ 0.88	96.03 $\pm$ 0.13	94.34 $\pm$ 0.79	97.21 $\pm$ 0.15	96.64 $\pm$ 0.50	96.11 $\pm$ 0.09	94.97 $\pm$ 0.15

Acc_u metrics (with values typically improving by 7%–8% across different datasets and tasks). When compared to other three incremental learning models (BLIP, R-DFCIL, and FACT), FAN-AFSIL also shows improvements of 2%–3% across different tasks. The significant improvement can be attributed to the FAN and AIC components proposed in the FAN-AFSIL model. FAN provides strong feature representation capabilities, while AIC enhances the model's ability to adapt to the current task (see Table 5).

Fig. 9 presents a comparison of the accuracy of the ten models mentioned above under varying levels of openness across four different datasets. The unknown attacks in the datasets are arranged in reverse order. In each dataset, we gradually add unknown attacks to the test set to adjust the level of openness. Performance is evaluated using the H_{score} , which provides a comprehensive assessment of the model's accuracy and recall. FAN-AFSIL consistently demonstrates exceptional performance across all levels of openness, particularly in the detection of IoT unknown attacks. Compared to other models, it achieves significantly higher H_{score} , effectively handling the challenges posed

by the introduction of IoT unknown attacks. This indicates that our model is not only robust in dealing with known attacks but also excels at identifying new, unseen attacks in an open-set environment. As the openness level increases, FAN-AFSIL continues to maintain its superiority, reinforcing its ability to adapt and perform well.

4.4. Ablation experiments and performance analysis

4.4.1. Comparison of feature visualization results

To further validate the effectiveness of the FAN-AFSIL model compared to other models, we use T-SNE to visualize the performance. The experiments are conducted under a 10-shot setting on four different datasets. Moreover, we present the visualizations of Known and Unknown attacks (as shown in Fig. 10), with the baseline being Transformer + FSIL. The results in Fig. 10 indicate that, with the continuous addition of optimization strategies, the FAN-AFSIL model exhibits the best visualization performance in distinguishing different IoT unknown intrusions. Compared to other models, the visualizations obtained by

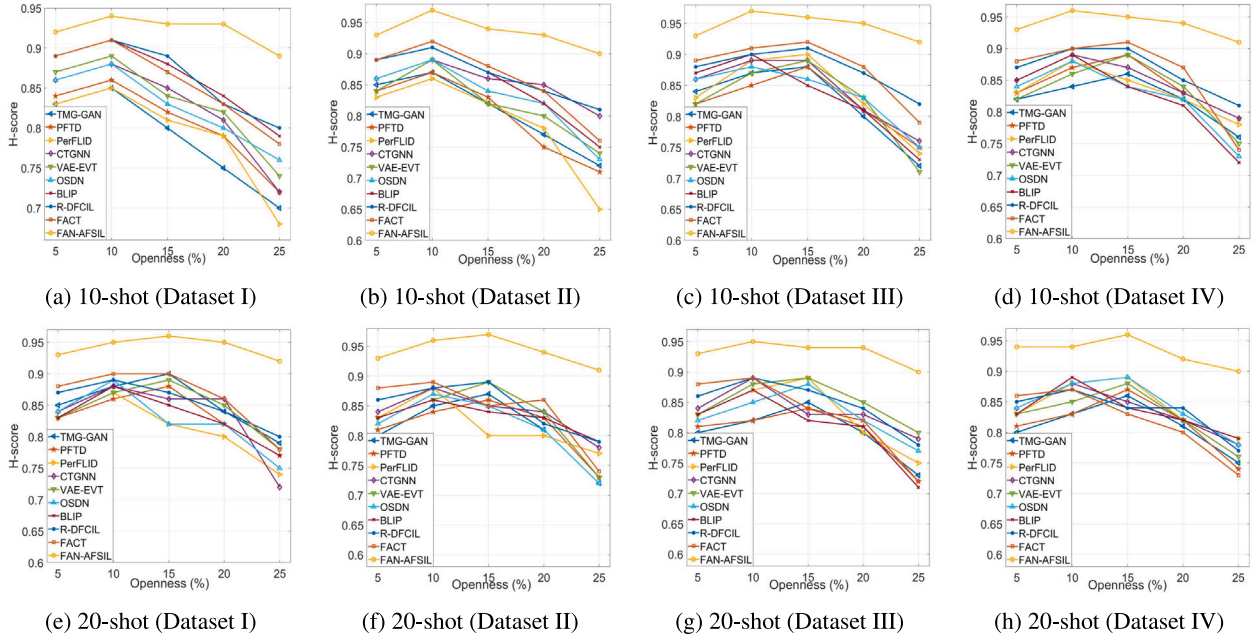


Fig. 9. Performance comparison under different openness levels.

Table 6

Performance of each SOTA model in detecting intrusions on different incremental tasks.

Model	Index	Dataset III				Dataset IV			
		Task I	Task II	Task III	Task IV	Task I	Task II	Task III	Task IV
TMG-GAN	Acc_k (%)↑	88.63 ± 1.21	87.89 ± 1.02	85.66 ± 0.47	82.58 ± 0.14	88.97 ± 0.25	87.86 ± 0.85	86.20 ± 1.09	83.58 ± 1.02
	Acc_u (%)↑	87.49 ± 0.38	86.11 ± 0.63	84.20 ± 1.26	82.03 ± 0.99	86.82 ± 0.46	85.30 ± 0.07	83.29 ± 0.54	80.55 ± 0.41
	H_{score} (%)↑	88.06 ± 0.58	86.99 ± 0.78	84.92 ± 0.68	82.30 ± 0.25	87.88 ± 0.32	86.56 ± 0.13	84.72 ± 0.72	82.04 ± 0.58
PFTD	Acc_k (%)↑	87.93 ± 0.06	86.57 ± 1.37	84.88 ± 1.06	81.55 ± 0.38	88.26 ± 0.57	87.19 ± 0.66	85.33 ± 0.07	82.74 ± 0.52
	Acc_u (%)↑	86.58 ± 0.05	85.86 ± 0.56	83.78 ± 0.44	80.93 ± 0.57	86.98 ± 0.53	85.83 ± 1.07	83.64 ± 0.71	81.78 ± 0.62
	H_{score} (%)↑	87.25 ± 0.05	86.21 ± 0.80	84.33 ± 0.62	81.24 ± 0.46	87.62 ± 0.55	86.50 ± 0.82	84.48 ± 0.13	82.26 ± 0.57
PerFLID	Acc_k (%)↑	90.42 ± 0.33	89.54 ± 0.41	87.80 ± 0.03	84.93 ± 0.58	90.04 ± 0.01	89.50 ± 0.45	87.21 ± 0.06	84.86 ± 0.15
	Acc_u (%)↑	88.97 ± 0.77	87.81 ± 0.30	85.86 ± 1.25	82.77 ± 1.04	89.83 ± 0.42	88.51 ± 0.93	86.89 ± 0.04	83.75 ± 0.39
	H_{score} (%)↑	89.69 ± 0.46	88.67 ± 0.35	86.82 ± 0.06	83.84 ± 0.74	89.93 ± 0.02	89.00 ± 0.61	87.05 ± 0.05	84.30 ± 0.22
CTGNN	Acc_k (%)↑	91.54 ± 0.09	90.76 ± 1.34	88.15 ± 0.36	85.77 ± 0.16	91.07 ± 0.50	90.63 ± 0.08	88.89 ± 0.44	86.03 ± 0.59
	Acc_u (%)↑	90.16 ± 1.28	89.48 ± 1.03	87.89 ± 0.67	85.02 ± 0.44	90.33 ± 1.04	89.41 ± 0.99	87.68 ± 0.07	84.87 ± 0.52
	H_{score} (%)↑	90.84 ± 0.17	90.12 ± 1.16	88.02 ± 0.47	85.39 ± 0.23	90.70 ± 0.68	90.02 ± 0.15	88.28 ± 0.12	85.45 ± 0.55
VAEEVT	Acc_k (%)↑	92.36 ± 0.04	91.89 ± 0.21	90.05 ± 0.78	88.06 ± 0.54	92.11 ± 0.39	91.75 ± 0.86	90.07 ± 0.42	87.93 ± 0.50
	Acc_u (%)↑	91.77 ± 1.06	90.52 ± 0.37	88.94 ± 0.55	85.81 ± 0.06	91.43 ± 1.24	90.88 ± 1.07	88.56 ± 1.15	85.41 ± 1.09
	H_{score} (%)↑	92.06 ± 0.08	91.20 ± 0.27	89.49 ± 0.65	86.92 ± 0.11	91.77 ± 0.59	91.31 ± 0.95	89.31 ± 0.62	86.65 ± 0.68
OSDN	Acc_k (%)↑	92.41 ± 0.56	92.02 ± 0.47	91.56 ± 0.78	89.89 ± 0.05	92.11 ± 0.03	91.75 ± 0.56	90.06 ± 0.04	88.89 ± 0.56
	Acc_u (%)↑	91.23 ± 0.37	90.14 ± 0.05	88.43 ± 0.62	86.01 ± 0.86	91.02 ± 0.57	90.66 ± 0.01	88.72 ± 0.53	86.14 ± 0.08
	H_{score} (%)↑	91.82 ± 0.45	91.07 ± 0.90	89.97 ± 0.69	87.91 ± 0.09	91.56 ± 0.06	91.20 ± 0.02	89.38 ± 0.07	87.49 ± 0.14
BLIP	Acc_k (%)↑	92.61 ± 1.10	92.03 ± 0.34	91.21 ± 0.07	89.33 ± 0.52	92.20 ± 0.64	91.76 ± 0.07	90.05 ± 1.24	87.62 ± 1.01
	Acc_u (%)↑	91.17 ± 0.06	90.85 ± 0.46	88.77 ± 0.24	85.86 ± 0.71	91.06 ± 0.55	90.65 ± 0.04	88.86 ± 0.27	85.93 ± 0.07
	H_{score} (%)↑	91.88 ± 0.11	91.44 ± 0.39	89.97 ± 0.11	87.56 ± 0.60	91.63 ± 0.59	91.20 ± 0.05	89.45 ± 0.44	86.77 ± 0.13
R-DFCIL	Acc_k (%)↑	93.58 ± 0.33	93.04 ± 0.48	92.15 ± 0.06	90.24 ± 0.54	93.71 ± 1.20	93.02 ± 0.75	92.55 ± 0.06	90.21 ± 0.08
	Acc_u (%)↑	92.30 ± 0.37	91.88 ± 0.06	90.02 ± 0.77	88.25 ± 0.03	92.12 ± 0.56	91.89 ± 0.48	89.63 ± 0.45	86.52 ± 0.01
	H_{score} (%)↑	92.94 ± 0.35	92.46 ± 0.11	91.07 ± 0.11	89.23 ± 0.06	92.91 ± 0.76	92.45 ± 0.59	91.07 ± 0.11	88.33 ± 0.02
FACT	Acc_k (%)↑	94.10 ± 0.03	93.84 ± 0.36	91.27 ± 0.81	89.03 ± 0.09	93.85 ± 0.42	93.11 ± 0.09	91.79 ± 0.32	89.24 ± 0.63
	Acc_u (%)↑	93.17 ± 0.06	92.65 ± 0.08	90.94 ± 0.06	87.88 ± 0.42	93.04 ± 0.47	92.99 ± 0.42	90.71 ± 0.09	87.63 ± 0.75
	H_{score} (%)↑	93.63 ± 0.04	93.24 ± 0.13	91.10 ± 0.11	88.45 ± 0.15	93.44 ± 0.44	93.05 ± 0.15	91.25 ± 0.14	88.43 ± 0.68
FAN-AFSIL	Acc_k (%)↑	97.25 ± 0.87	96.84 ± 0.01	96.17 ± 0.96	94.88 ± 0.04	97.37 ± 0.58	96.48 ± 0.09	95.87 ± 0.03	95.08 ± 0.66
	Acc_u (%)↑	96.31 ± 0.06	96.02 ± 1.10	95.69 ± 0.37	95.14 ± 0.08	96.20 ± 0.71	95.44 ± 0.15	95.10 ± 0.78	94.96 ± 1.37
	H_{score} (%)↑	96.78 ± 0.11	96.43 ± 0.02	95.93 ± 0.53	95.01 ± 0.05	96.78 ± 0.64	95.96 ± 0.11	95.48 ± 0.06	95.02 ± 0.89

FAN-AFSIL show no significant overlap or confusion in distinguishing between the different known and unknown categories. Furthermore, the resulting visualizations are notably compact, demonstrating strong category separability. Therefore, FAN-AFSIL achieves the best results in recognizing different IoT unknown intrusions. When PES is introduced, the model performance across different datasets is further enhanced.

The results show that AIC effectively models global task correlations, facilitates the effective integration of the unified classifier, and adapts the feature representations of test samples using global task information, making them better aligned with the current classification task, thereby achieving the best visualization results.

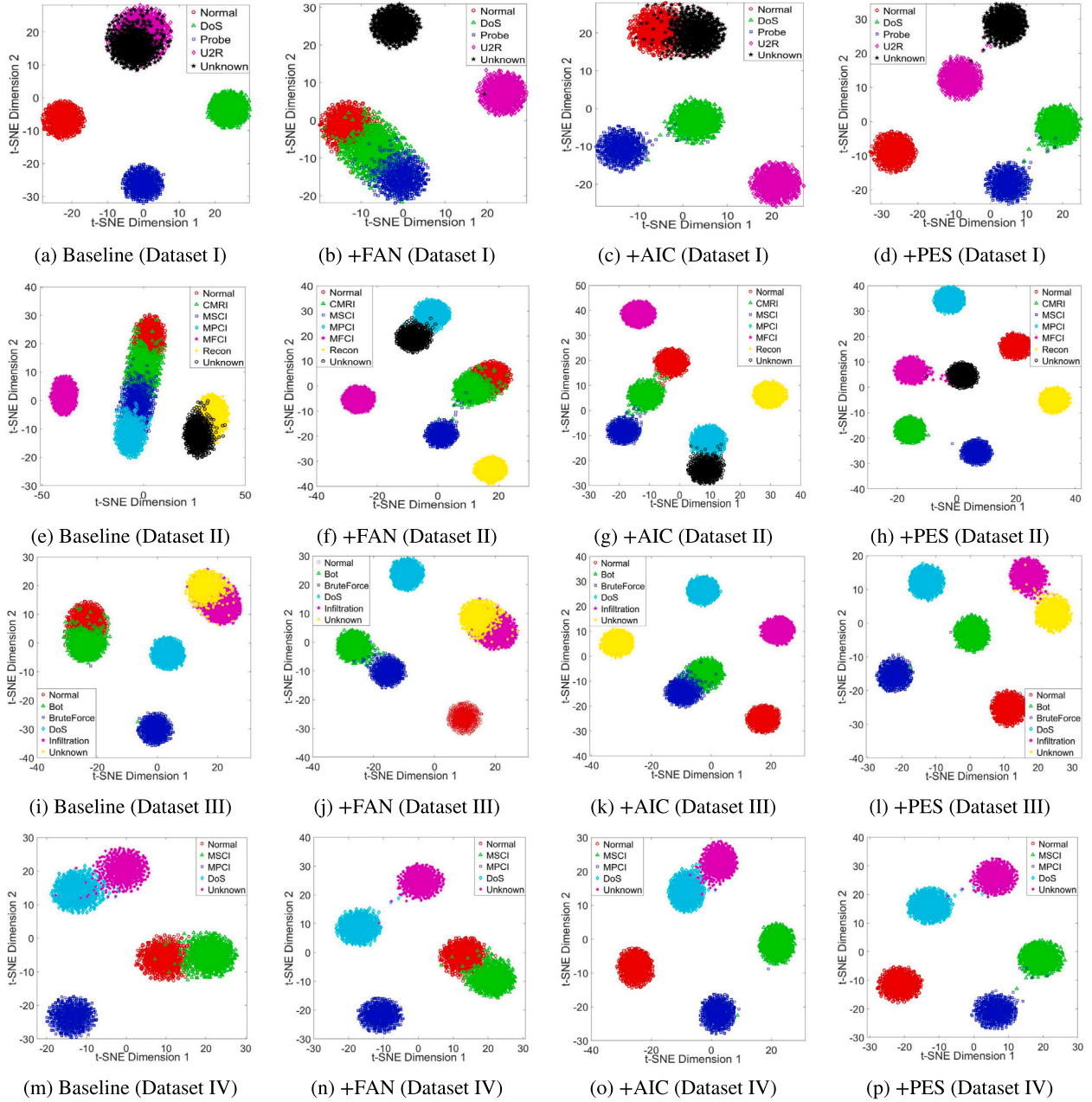


Fig. 10. Feature visualization results for different models.

4.4.2. Module ablation analysis

To quantitatively assess the individual contributions of each module, we conduct open-set detection experiments under four incremental tasks using Dataset IV. The baseline model used is Transformer + FSIL. We present the performance metrics of the FAN-AFSIL model as strategies are progressively introduced, with results shown in Tables 7–8. Based on the results in Tables 7–8, the following conclusions can be drawn: consistent with the visualization findings in Fig. 10, the performance metrics of FAN-AFSIL across different incremental tasks show a clear upward trend as each module is gradually incorporated. When using only the baseline model, the performance under open-set conditions is relatively limited. With the sequential integration of key modules such as FAN and AIC, the model's ability to distinguish between known and unknown attacks is significantly enhanced. After the final PES strategy, FAN-AFSIL achieves the best performance across all four incremental detection tasks, outperforming configurations where certain modules are not yet integrated. These

results strongly validate the effectiveness of the core components within FAN-AFSIL and demonstrate that the proposed strategies play a crucial role in improving model generalization and enhancing robustness in detecting IoT unknown attacks.

4.5. Verification of strategy effectiveness

4.5.1. Dual res-transformer performance verification

Since the FAN module in the FAN-AFSIL model is based on our proposed Dual Res-Transformer, we conduct a detailed analysis to explore the necessity and effectiveness of the strategies incorporated within the Dual Res-Transformer. Using T-SNE, we visualize the feature distributions as various strategies are progressively introduced. In this process, the classifier used is our proposed AIC classifier. To further validate the effectiveness of the Dual Res-Transformer, we illustrate the distribution of different attack categories on Dataset IV, demonstrating the necessity of each strategy. Additionally, we label three

Table 7
Ablation results on incremental Tasks I–II.

Baseline	FAN	AIC	PES	Task I			Task II		
				$Acc_k/(%) \uparrow$	$Acc_u/(%) \uparrow$	$H_{score}/(%) \uparrow$	$Acc_k/(%) \uparrow$	$Acc_u/(%) \uparrow$	$H_{score}/(%) \uparrow$
✓	–	–	–	84.16 ± 1.36	83.91 ± 0.24	84.03 ± 0.41	83.74 ± 1.02	83.20 ± 0.39	83.47 ± 0.56
✓	✓	–	–	90.94 ± 0.45	90.26 ± 0.08	90.60 ± 0.14	90.30 ± 0.08	90.07 ± 0.55	90.18 ± 0.14
✓	✓	✓	–	93.13 ± 0.79	92.97 ± 0.39	93.05 ± 0.52	92.86 ± 0.04	92.25 ± 0.84	92.55 ± 0.08
✓	✓	✓	✓	95.84 ± 0.05	95.38 ± 0.42	95.61 ± 0.09	95.31 ± 0.69	95.06 ± 0.07	95.18 ± 0.13

Table 8
Ablation results on incremental Tasks III–IV.

Baseline	FAN	AIC	PES	Task III			Task IV		
				$Acc_k/(%) \uparrow$	$Acc_u/(%) \uparrow$	$H_{score}/(%) \uparrow$	$Acc_k/(%) \uparrow$	$Acc_u/(%) \uparrow$	$H_{score}/(%) \uparrow$
✓	–	–	–	85.94 ± 0.01	84.25 ± 0.77	85.09 ± 0.02	84.71 ± 0.45	83.89 ± 0.03	84.30 ± 0.06
✓	✓	–	–	89.41 ± 0.59	88.84 ± 1.02	89.12 ± 0.78	91.69 ± 0.37	91.12 ± 1.05	91.40 ± 0.55
✓	✓	✓	–	94.57 ± 0.15	93.16 ± 0.33	93.86 ± 0.21	93.55 ± 0.08	93.20 ± 0.79	93.37 ± 0.15
✓	✓	✓	✓	97.78 ± 0.53	96.62 ± 0.08	97.20 ± 0.14	96.61 ± 0.44	95.93 ± 0.64	96.27 ± 0.52

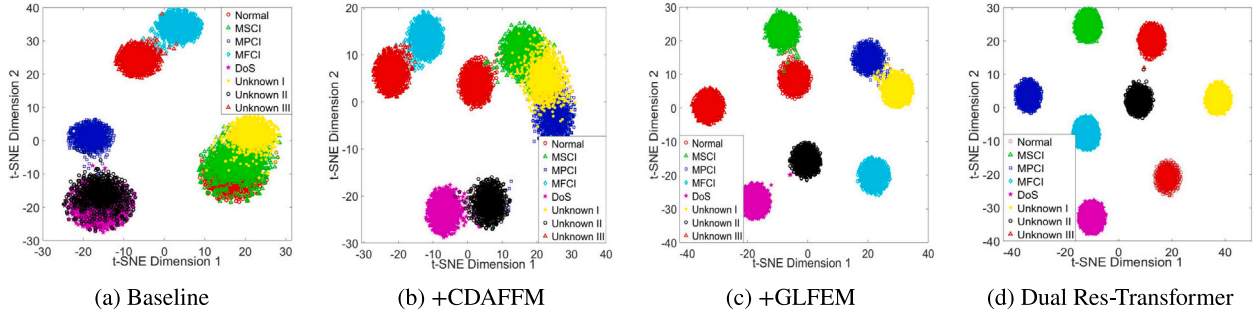


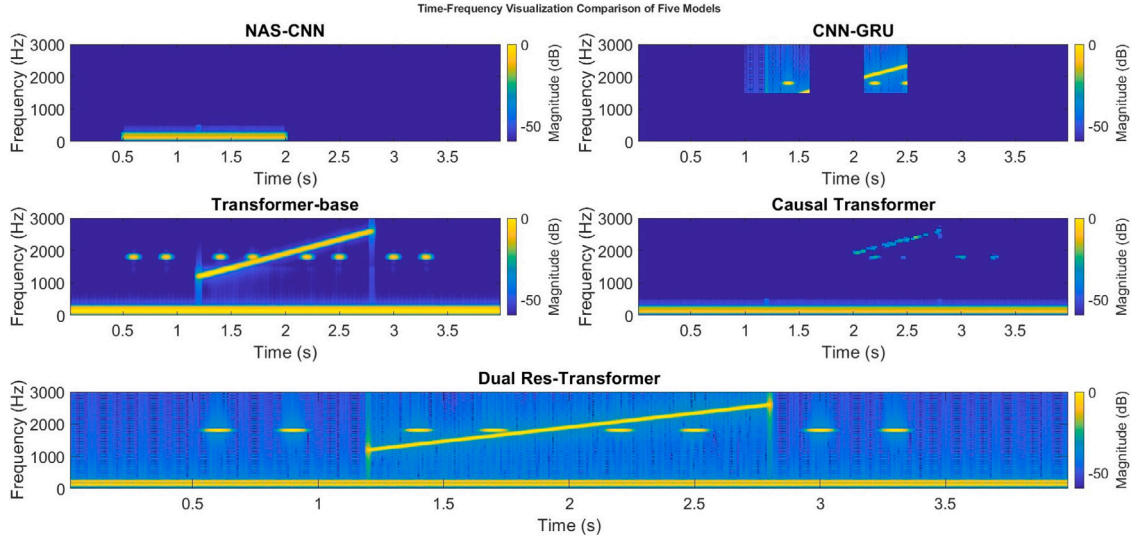
Fig. 11. Dual Res-Transformer performance verification.

unknown categories in Dataset IV as Unknown I–III to more comprehensively evaluate the feasibility of Dual Res-Transformer in open-set scenarios. As shown in Fig. 11, with the gradual integration of each strategy in the Dual Res-Transformer, the quality of the feature visualizations improves significantly, especially with a noticeable reduction in the overlap between unknown categories. These results verify the feasibility and effectiveness of each component within the Dual Res-Transformer. Specifically, the designed GADRM significantly enhances the model ability to extract meaningful features from attack patterns. Furthermore, the proposed GLFEM, which integrates the strengths of both CNNs and Transformers, allows the network to effectively capture both local details and global semantic context. This design enables the model to extract richer, high-level semantic features, thereby further improving its overall feature representation and discriminative capability.

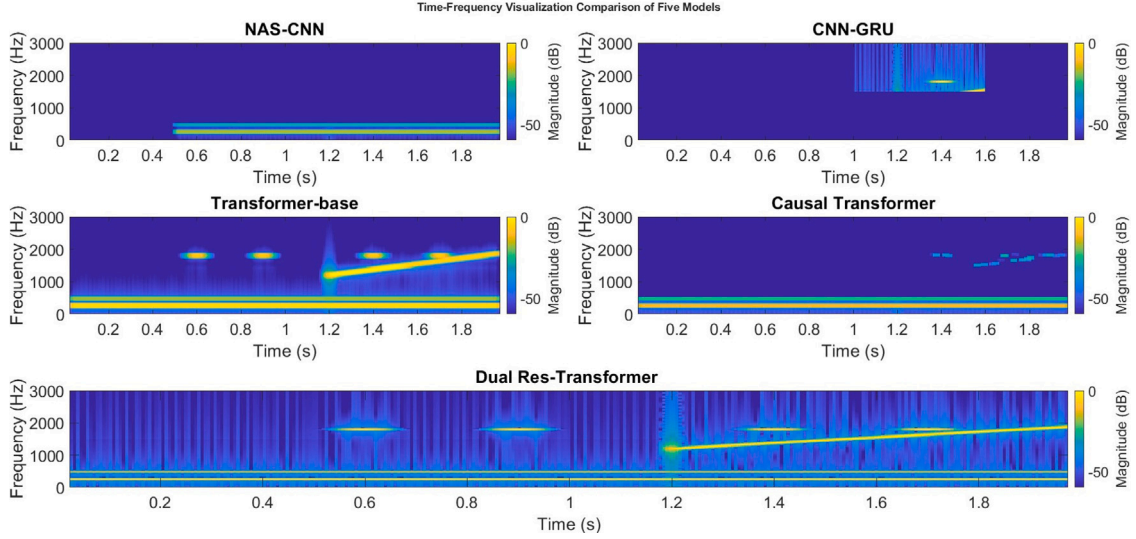
To comprehensively evaluate the feature learning performance of the proposed Dual Res-Transformer in IoT attack detection, four representative models: NAS-CNN (Lu et al., 2025), CNN-GRU (Hu et al., 2025), Transformer-base (Wu et al., 2025), and Causal Transformer (Zeng et al., 2025) are selected for comparative feature visualization experiments based on Dataset II and Dataset IV (as shown in Fig. 12). The results demonstrate that the proposed Dual Res-Transformer can stably and completely capture the key characteristics of attack samples throughout the entire time sequence. It not only identifies the global variation trends of attack behaviors but also accurately captures high-frequency detailed information, producing clearer and more distinguishable feature representations. This superior performance can be attributed to the collaborative effect of its dual residual connections and multi-scale attention fusion mechanism, which enable a balanced representation between global semantics and local high-frequency perception. In contrast, CNN-based models (NAS-CNN and CNN-GRU) exhibit limitations in handling complex time–frequency features, failing to effectively preserve temporal correlations and high-frequency details.

Transformer-based models, while capable of global modeling, tend to produce blurred high-frequency features with indistinct boundaries. Overall, the Dual Res-Transformer outperforms the comparative models in terms of feature comprehensiveness, detail sensitivity, and representation clarity, thereby validating its effectiveness and robustness in IoT attack feature learning and detection tasks.

To validate the efficiency of the FAN model, we conducted comparative experiments with baseline models (CNN, LSTM, and Transformer) and two FAN variants (without CDAFFM and without GLFEM) on Datasets I–IV under an openness level of 5%. Tables 9–12 report the average accuracy, variance, parameter size, training time, and memory consumption over 10 epochs. To ensure a fair comparison, all models adopted the same data split, batch size, optimizer, learning rate schedule, hardware environment, and evaluation metrics. The batch size was set to 32, and all models were trained using SGD. For Datasets I–II, the initial learning rate was 0.01 and was reduced after the 30th and 60th epochs. For Datasets III–IV, the initial learning rate was 0.1 and was reduced after the 30th and 40th epochs. Although architecture-specific parameters varied across models, the overall training and evaluation conditions were kept consistent. The results show that traditional models achieve lower accuracy and higher variance; for example, on Dataset I, CNN attains 82.639% accuracy with 0.8861% variance, while Transformer reaches 90.364% with 0.7894% variance, whereas FAN achieves 96.506% accuracy with a significantly reduced variance of 0.3042%. Similar trends are observed across Datasets II–IV, where FAN consistently maintains the highest accuracy (95.137%, 94.268%, and 93.517%) and the lowest variance, demonstrating superior stability and robustness. Although the two FAN variants outperform conventional baselines, they still underperform compared to the others, confirming that both CDAFFM and GLFEM are essential: CDAFFM enhances feature fusion via channel attention, while GLFEM improves feature representation by capturing global and local information. While FAN incurs slightly higher computational cost (in parameters, time, and



(a) Feature visualization performance on Dataset II



(b) Feature visualization performance on Dataset IV

Fig. 12. Feature learning performance verification.

Table 9

Base model comparison in Dataset I (openness=5%).

Model	Accuracy /%	Variance /%	Parameters /M	Time /S	Memory /MB
CNN	82.639	0.8861	79.110	29.63	51.368
LSTM	84.218	0.9152	75.007	28.44	55.610
Trans	90.364	0.7894	90.004	34.38	60.304
ViT	91.785	0.7416	94.697	36.91	65.118
Vari-I	93.673	0.5510	90.224	38.22	69.307
Vari-II	93.771	0.5126	91.628	39.07	70.245
our	96.506	0.3042	96.229	40.11	71.251

Table 10

Base model comparison in Dataset II (openness=5%).

Model	Accuracy /%	Variance /%	Parameters /M	Time /S	Memory /MB
CNN	80.912	0.9324	78.563	28.74	50.882
LSTM	82.547	0.9018	74.321	27.96	54.973
Trans	88.736	0.8127	89.447	33.12	59.428
ViT	90.264	0.7693	93.882	35.67	63.907
Vari-I	92.118	0.5986	89.903	36.95	67.812
Vari-II	92.406	0.5612	91.174	37.88	68.955
our	95.137	0.3379	95.608	39.02	70.463

memory) than simpler models, this overhead is moderate. The gains in accuracy and stability are substantial, indicating a better trade-off between performance and efficiency for open-set intrusion detection.

4.5.2. Performance validation of PES strategy

To validate the necessity of the PES strategy in FAN-AFSIL, we conduct tests on incremental tasks and varying levels of openness involving both known and unknown attacks. We evaluate the performance of the FAN-AFSIL model with and without the PES strategy, using Acc_k

and Acc_u . The experimental results, shown in Fig. 13, indicate that when the PES strategy is included in the FAN-AFSIL model, both Acc_k and Acc_u metrics perform best across the four datasets. After adding the PES strategy, the model's metric values in different incremental tasks consistently remain above 90%, significantly outperforming the results when PES are not included. We also observe that the FAN-AFSIL model demonstrates strong performance across different levels of openness when incorporating the PES strategy. This further confirms the effectiveness of the FAN-AFSIL model in detecting IoT

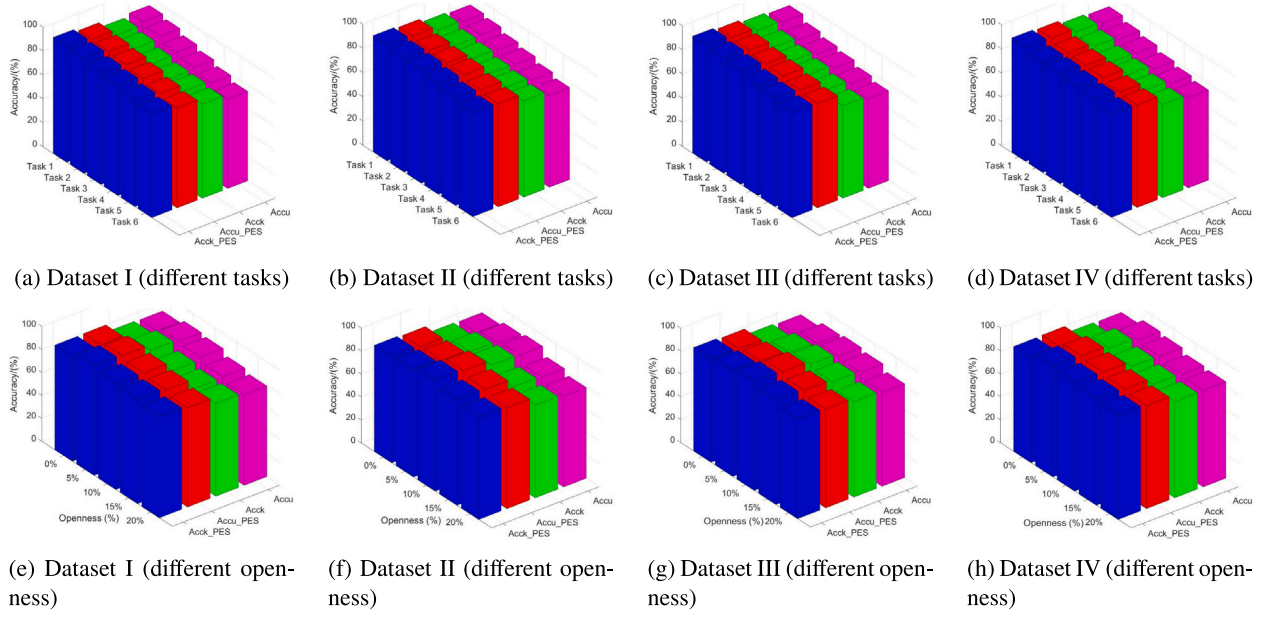


Fig. 13. Performance validation of PES strategy.

Table 11

Base model comparison in Dataset III (openness=5%).

Model	Accuracy /%	Variance /%	Parameters /M	Time /S	Memory /MB
CNN	79.384	0.9687	77.942	27.85	50.116
LSTM	81.063	0.9345	73.588	27.21	53.762
Trans	87.295	0.8459	88.763	32.47	58.903
ViT	88.912	0.8026	92.975	34.92	62.871
Vari-I	91.447	0.6328	89.118	36.18	66.524
Vari-II	91.782	0.6014	90.507	37.05	67.733
Our	94.268	0.3651	94.836	38.26	69.284

Table 12

Base model comparison in Dataset IV (openness=5%).

Model	Accuracy /%	Variance /%	Parameters /M	Time /S	Memory /MB
CNN	78.527	0.9953	77.316	26.93	49.604
LSTM	80.148	0.9621	72.884	26.35	53.118
Trans	86.742	0.8735	88.092	31.84	58.227
ViT	88.109	0.8297	92.214	34.16	61.945
Vari-I	90.836	0.6684	88.447	35.42	65.733
Vari-II	91.203	0.6298	89.836	36.27	66.982
Our	95.517	0.3926	94.105	37.63	68.541

attacks within an open-set environment. This improvement is attributed to the PES strategy, which generates a series of pseudo-incremental tasks through data mixing and trains the model using meta-learning. This process enables the model to learn generalized features between different pseudo-incremental tasks, thus exhibiting stronger detection capabilities.

4.5.3. Performance verification on real and pseudo-incremental package tasks

To further verify the effectiveness of the proposed model in analyzing incremental tasks, we employed the t-SNE method to visualize and compare the distribution of three types of samples (original base-category samples, pseudo-samples generated by the PES strategy, and new attack samples from real incremental tasks) in the high-dimensional feature space. As shown in Fig. 14, without the PES strategy, the distributions of pseudo-incremental and real samples exhibit significant differences in the embedding space, characterized by

scattered clustering and blurred boundaries. After introducing the PES strategy, the feature distributions of pseudo and real samples become more aligned, with clearer cluster boundaries and a notably reduced inter-domain gap. Further statistical analysis indicates that incorporating the PES strategy enhances the separability of feature distributions across the four datasets. These results demonstrate that the PES strategy effectively mitigates the domain shift between pseudo-incremental and real incremental tasks, enabling the model to maintain stable feature extraction performance.

4.5.4. Performance verification on HRP

To further evaluate the performance of HRP, we adopt two key metrics (plasticity and forgetting) to assess the adaptability in evolving attack scenarios. The new attack categories are introduced, with forgetting measuring the performance degradation on previously known categories, and plasticity reflecting the model's ability to effectively learn new categories. To further verify the feasibility of the proposed HRP model, we present the results when the proposed method includes and does not include this strategy. As shown in Fig. 15, the 2D plots illustrate performance trends, while the 3D contour plots and trajectories highlight the evolution process over time. Using Dataset I as an example, when containing HRP strategy, it consistently demonstrates low forgetting while improving the recognition of new categories across different data distributions. In contrast, models without HRP experience high forgetting and show limited plasticity as new categories are introduced. Despite minor fluctuations, our model remains stable throughout, showcasing its robustness and effectiveness in detecting unknown intrusions in IoT environments.

4.5.5. Classification decision experiment under continuously evolving attacks

In this section, we conduct a classification decision experiment under continuously evolving attacks. The new attack categories are progressively introduced on Datasets II and IV to examine the separability between known and unknown categories, as illustrated in Fig. 16. Taking Dataset II as an example, the known categories form compact and well-separated clusters at the initial stage. Moreover, unknown samples are distributed in low-density peripheral regions, indicating a promising initial open-set recognition capability. As new categories are gradually introduced, the emerging categories quickly form distinct clusters with smooth centroid transitions, suggesting that the proposed

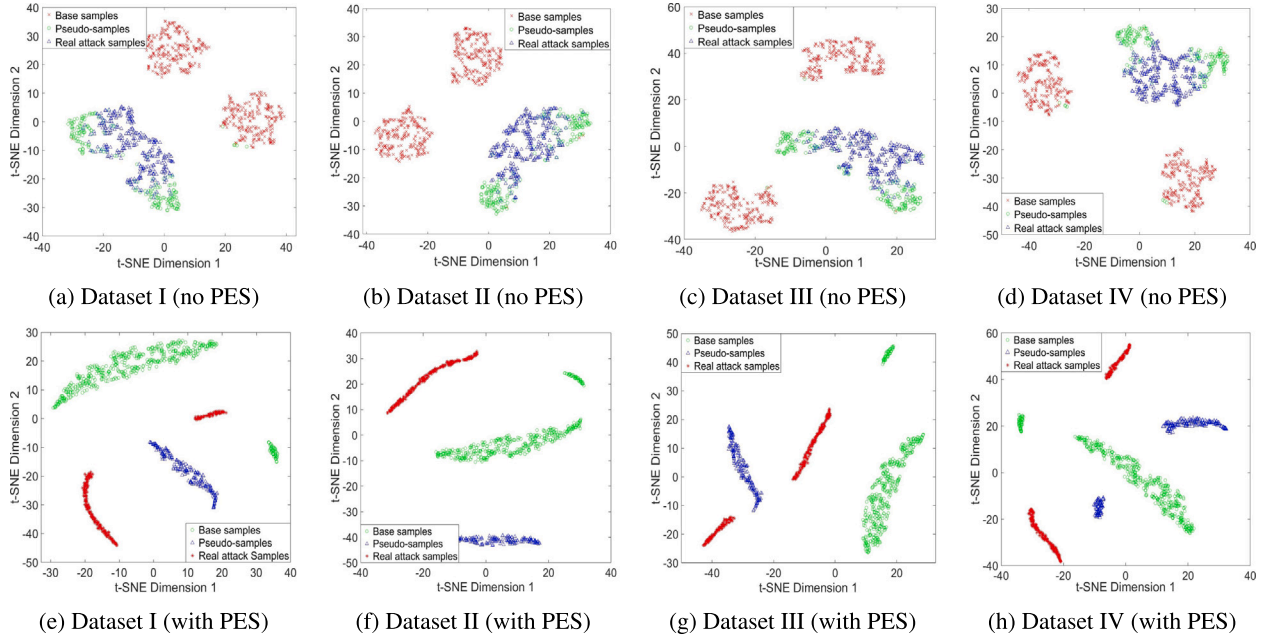


Fig. 14. Performance verification on real and pseudo-incremental tasks.

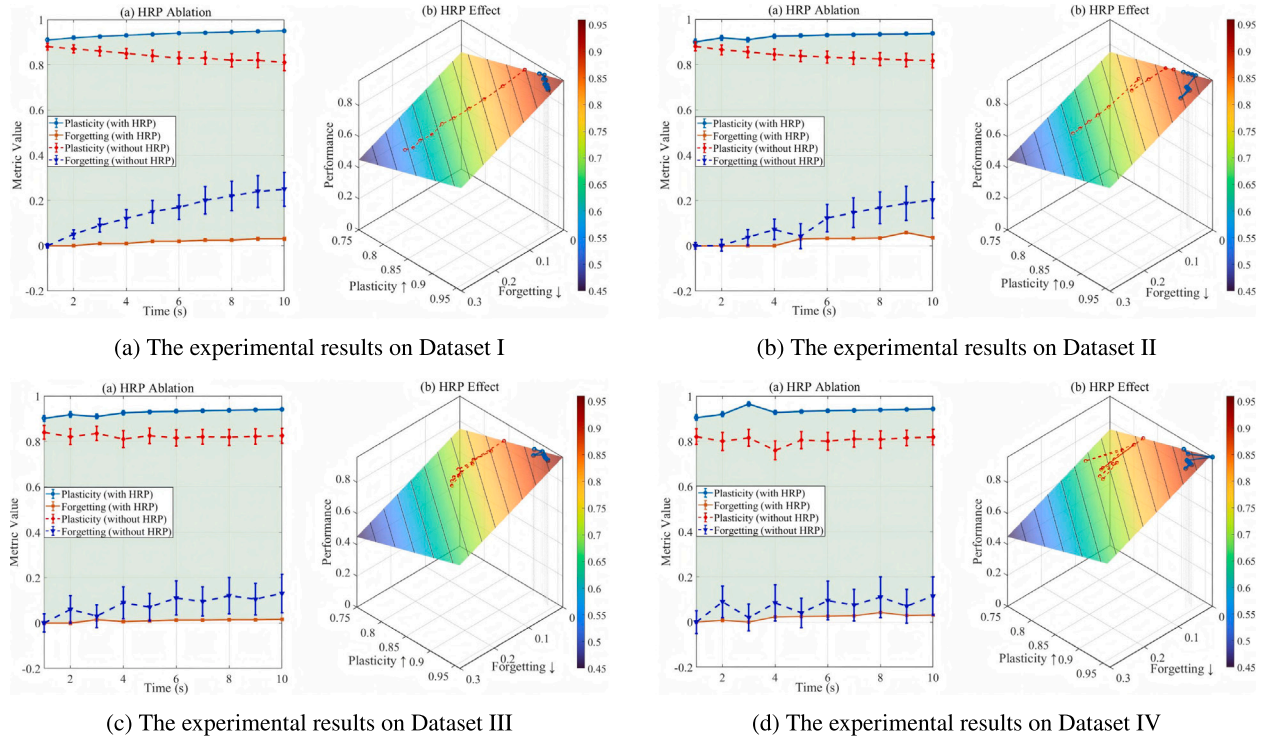


Fig. 15. Performance verification on HRP.

model effectively alleviates catastrophic forgetting during incremental learning. Even as the number of categories continues to increase, the decision boundaries remain clear and non-overlapping, and unknown samples are consistently separated from known categories. Similarly, Fig. 16(b) shows that, under the more complex setting of Dataset IV with multiple coexisting attack categories, the feature space remains well organized, and unknown attacks are still distributed in distinct regions. These results indicate that freezing FAN does not impair adaptation to later tasks. Instead, by combining a stable generalized feature space with HRP-based relational adjustment, the proposed framework

can continuously accommodate new attack categories while preserving discrimination for known and unknown categories.

4.6. Parameter sensitivity analysis

To analyze the impact of the Number of query set size (E) and the initial learning rate on the performance of the FAN-AFSIL model, we present the performance across different incremental tasks under various values, as shown in Fig. 17. The experimental results indicate that when the query set size (E) is uniformly set to 10, the FAN-AFSIL model

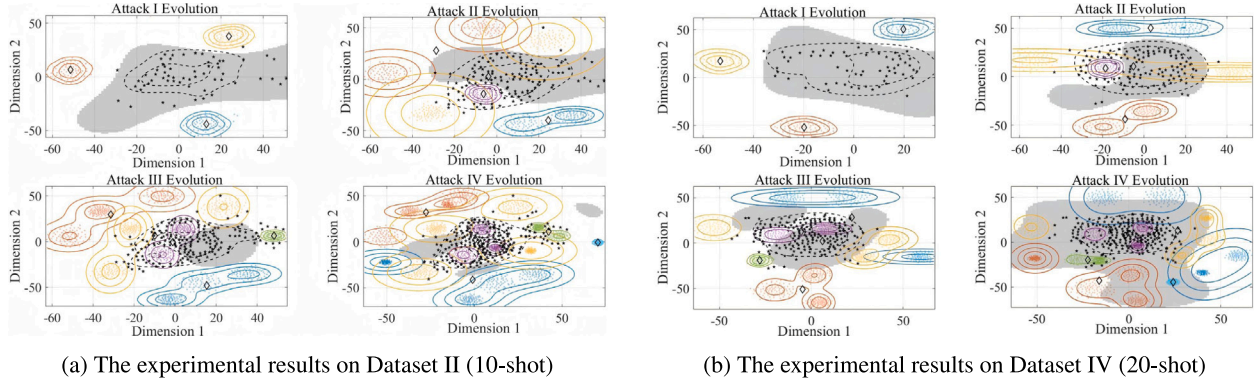


Fig. 16. Performance verification on decision under continuously evolving attacks.

achieves the best performance across all four datasets. This suggests that an appropriate query set size is crucial for the adaptability in incremental learning tasks. If the value of E is too large or too small, it may degrade the transferability between tasks, thus affecting performance. Regarding the impact of the initial learning rate, the results show that its optimal value varies across different datasets. In Dataset I, the FAN-AFSIL model performs best when the initial learning rate is 0.0002, indicating that this learning rate achieves good convergence speed and stability for this dataset. Other datasets exhibit different optimal learning rates: for Dataset II, the best performance is achieved with an initial learning rate of 0.0005; for Dataset III, the optimal learning rate is 0.001; and for Dataset IV, a learning rate of 0.0008 yields the best performance. These results suggest that the initial learning rate should be adjusted according to the characteristics of the dataset, as both too high and too low learning rates can negatively impact the model's training and convergence process. To evaluate the impact of the mixing coefficient λ in the pseudo-task selection (PES) process, we constructed pseudo-incremental samples using the formulation $I_{ab} = \lambda I_a + (1 - \lambda) I_b$, where $\lambda \in [0.2, 0.8]$, as well as values drawn from the Beta distribution $\lambda \sim \text{Beta}(\alpha, \alpha)$ with $\alpha \in \{0.2, 0.4, 1.0, 2.0\}$. All experiments are conducted under identical training conditions. We recorded the curves of the model's average accuracy on real new categories (incremental tasks) across different stages, as shown in Fig. 17(i)–17(l). The experimental results indicate that when λ takes extreme values (close to 0 or 1), the generated pseudo-tasks exhibit low diversity, leading to reduced generalization ability of the model. In contrast, moderate random mixing yields a more balanced feature distribution. Notably, when λ follows the Beta (0.4, 0.4) distribution, the model achieves the optimal balance between task representativeness and training stability. This distribution generates samples that are both similar to original categories, thereby expanding category boundaries while maintaining feature consistency. These findings verify that a moderately stochastic λ distribution enables the PES mechanism to produce more representative pseudo-incremental tasks, significantly enhancing the robustness and adaptability of the HRP model in continual learning scenarios.

4.7. Analysis of anti-disturbance performance

To systematically evaluate the robustness of the model against adversarial perturbations, we designed multi-level perturbation experiments on four datasets (Dataset I–IV) and plotted the corresponding robustness curves (Fig. 18). Controlled perturbations were applied in the traffic space: in the feature space, channel mixing and amplitude scaling operations were employed. The perturbation intensity was set to $\epsilon \in \{0, 0.5\%, 1\%, 2\%, 4\%\}$. All models were trained under clean (non-perturbed) conditions, and perturbations were injected only during the testing phase. The open-set detection accuracy is used as the primary evaluation metric. We can see that, as the perturbation intensity increases, the accuracy of the FAN-AFSIL model decreases; however, the

decrease is not significant. Meanwhile, we found that under different levels of disturbances, the performance of the FAN-AFSIL model does not change much on various datasets. Therefore, this indicates that the FAN-AFSIL model can stably perform Internet of Things intrusion detection in the presence of severe adversarial disturbances. Even when $\epsilon = 4\%$, FAN-AFSIL maintains over 85% detection accuracy. These findings provide supplementary evidence that the overall incremental framework remains robust under perturbations, which is consistent with the stability of the proposed prototype update mechanism.

4.8. Complexity analysis and evaluation

In this section, we evaluate the complexity of the proposed model for IoT attack detection in open-set environments from multiple perspectives. First, we select training time, test time, and memory consumption (MB) as the metrics, with the results shown in Table 14. As shown, FAN-AFSIL achieves lower complexity metrics than most methods across different datasets. Although, on some datasets, the training time of FAN-AFSIL is slightly higher than that of BLIP and R-DFCIL, the differences remain limited and within an acceptable range. This indicates that FAN-AFSIL can effectively detect unknown IoT attacks with relatively low memory usage and time consumption. Consequently, the computational overhead of prototype maintenance and relational modeling may also increase, especially in PSP and FCP where prototype interactions are established through attention-based mechanisms. Therefore, scalability may become a potential concern in large-scale class expansion scenarios.

Nevertheless, the results in Table 13 show that, under the current open-set IoT intrusion detection setting, the additional overhead introduced by the prototype-based framework remains manageable in terms of time and memory. Next, we further adopt model parameters, FLOPs (floating-point operations), and perplexity (PPL) as additional complexity metrics, where lower values indicate better performance. Complexity evaluations are conducted under different incremental tasks and open-set environments on Dataset IV, with the results shown in Tables 14–15. From Table 14, FAN-AFSIL achieves the best performance in Task I with Parameters/60.334M, FLOPs/71.226G, and PPL/65.883, while maintaining competitive performance across the remaining tasks. From Table 15, under the openness = 5% setting, FAN-AFSIL obtains the best performance with Parameters/59.674M, FLOPs/70.200G, and PPL/63.776, and continues to perform consistently well in other open-set settings.

To assess the real-time deployment capability of FAN-AFSIL in resource-constrained IoT environments, we conducted a latency analysis across four datasets (NSL-KDD, Gas Pipeline, CICIDS2017, and Water Storage Tank) under openness levels of 5%, 10%, and 15%. All latency experiments were conducted under the same hardware and software environment to ensure the fairness and consistency of comparisons among different methods. Specifically, the experiments

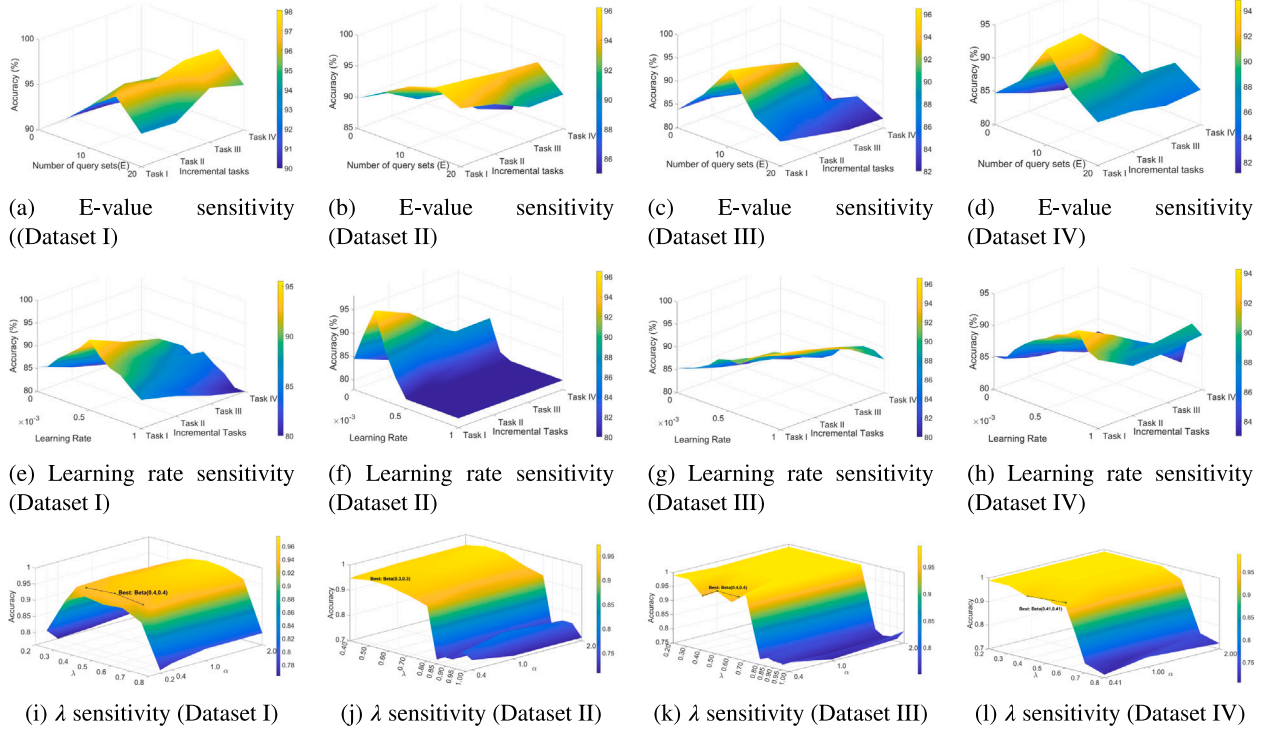


Fig. 17. Parameter sensitivity analysis.

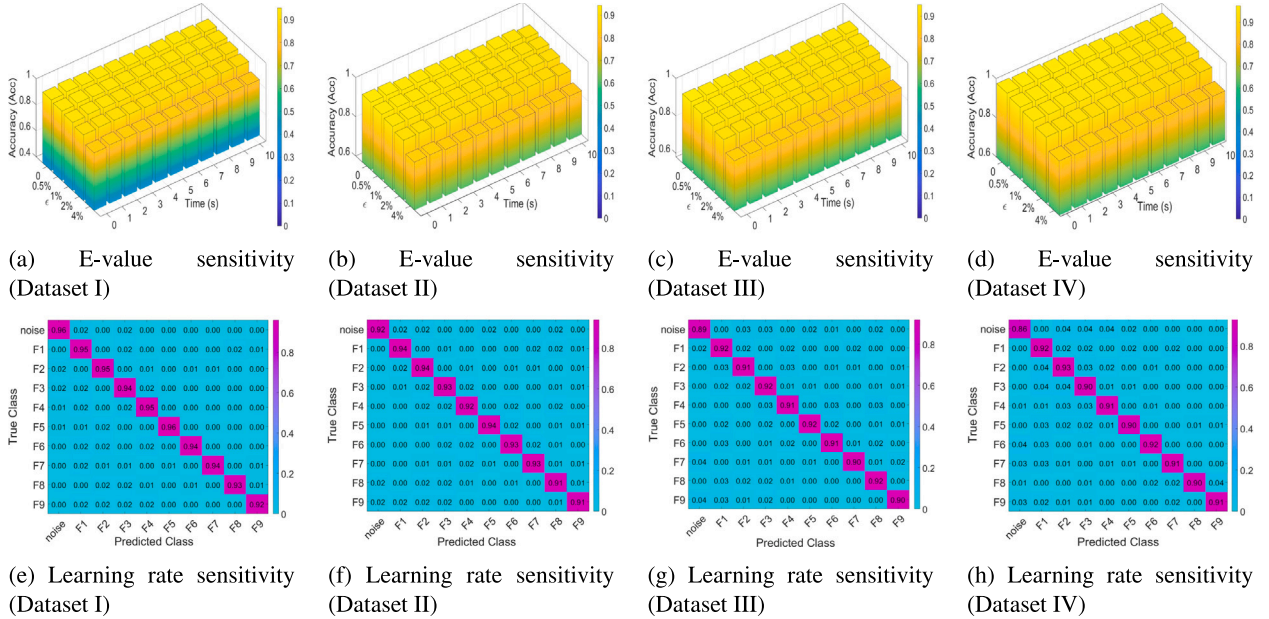


Fig. 18. Analysis of anti-disturbance performance.

were performed on a workstation equipped with an Intel Xeon W-2133 CPU, 64 GB RAM, and an NVIDIA Quadro RTX 5000 GPU with 16 GB memory. The operating system was Windows, and all models were implemented using Python and the PyTorch deep learning framework. For each method, latency evaluation was conducted during the inference stage after model training had been completed, and the average inference time per sample (ms/sample) was adopted as the evaluation metric for practical inference efficiency. As shown in Table 16, FAN-AFSIL consistently achieves the lowest latency across all settings; for example, on Dataset I, its latency increases moderately from

8.624 ms (5%) to 11.631 ms (10%) and 15.872 ms (15%), remaining significantly lower than baseline methods such as TM-GAN (20.336 ms $\rightarrow$ 26.782 ms $\rightarrow$ 35.914 ms). Similar patterns are observed across Datasets II–IV, where FAN-AFSIL maintains both low absolute latency and a smaller growth rate as openness increases. Although latency rises slightly with task complexity, it remains within an acceptable range, demonstrating that FAN-AFSIL achieves an effective balance between detection performance and real-time responsiveness, making it suitable for deployment in resource-constrained IoT environments. These results indicate that, despite the expansion of the prototype

Table 13
Complexity analysis on different datasets.

Model	Dataset I			Dataset II			Dataset III			Dataset IV		
	Training /s	Test /s	Memory /MB	Training /s	Test /s	Memory /MB	Training /s	Test /s	Memory /MB	Training /s	Test /s	Memory /MB
TM-GAN	59.634	11.267	85.645	60.245	11.785	85.781	58.635	11.002	84.631	61.389	11.996	86.347
PFTD	62.318	10.359	110.478	63.578	10.851	110.662	61.547	10.114	109.652	64.525	10.978	111.321
PerFLID	70.259	12.633	100.245	71.204	12.924	100.510	69.610	12.205	99.624	72.006	13.123	101.267
CTGNN	57.645	9.148	102.367	58.053	9.445	102.526	56.445	9.004	101.367	59.374	9.772	103.129
VAEEVT	50.367	7.659	89.645	51.668	7.828	89.678	49.637	7.365	88.962	52.118	7.965	90.204
OSDN	61.202	11.964	98.612	62.979	12.347	98.643	60.258	11.741	97.844	63.097	12.581	99.643
BLIP	41.275	8.675	88.279	42.355	8.995	88.526	40.200	8.338	87.593	43.471	9.240	89.105
R-DFCIL	40.929	7.322	65.126	41.571	7.659	65.397	40.911	7.106	64.228	42.226	7.867	66.315
FACT	45.396	6.881	59.140	46.285	6.999	59.428	44.353	6.452	58.106	47.413	7.110	60.207
Our	42.314	5.754	55.333	42.863	5.861	55.756	41.674	5.363	54.619	43.115	5.945	56.101

Table 14
Complexity analysis on different incremental tasks.

Model	Task I			Task II			Task III			Task IV		
	Parame_ ters /M	FLOPs /G	PPL	Parame_ ters /M	FLOPs /G	PPL	Parame_ ters /M	FLOPs /G	PPL	Parame_ ters /M	FLOPs /G	PPL
TM-GAN	89.645	69.317	63.117	90.345	71.245	64.511	92.417	73.692	66.587	95.632	76.125	69.110
PFTD	92.356	90.255	75.149	93.451	91.631	76.527	94.608	93.677	78.964	97.751	96.517	81.574
PerFLID	87.414	88.620	90.633	88.752	89.610	91.362	89.409	91.259	93.618	92.549	94.631	96.528
CTGNN	82.331	86.723	88.441	83.694	87.442	89.103	84.551	89.631	91.220	87.614	92.554	94.606
VAEEVT	67.459	87.914	71.620	68.542	88.636	72.681	70.234	90.115	74.505	73.525	93.781	77.551
OSDN	66.357	95.329	99.005	67.828	96.645	98.227	69.326	111.682	102.336	72.138	116.277	105.634
BLIP	71.202	98.651	84.291	72.447	99.613	85.363	74.851	101.369	87.961	77.861	104.214	89.912
R-DFCIL	61.075	86.410	90.227	62.586	87.698	91.455	64.274	89.454	93.607	67.889	92.509	95.663
FACT	68.945	70.162	89.636	69.111	71.054	90.568	71.663	73.610	93.005	74.611	76.718	95.977
Our	60.334	71.226	65.883	60.610	71.693	65.912	62.874	72.105	66.304	63.983	73.910	67.841

Table 15
Complexity analysis on different open environments.

Model	openness=5%			openness=10%			openness=15%			openness=20%		
	Parame_ ters /M	FLOPs /G	PPL	Parame_ ters /M	FLOPs /G	PPL	Parame_ ters /M	FLOPs /G	PPL	Parame_ ters /M	FLOPs /G	PPL
TM-GAN	86.941	67.541	61.258	88.741	68.912	62.378	90.429	70.615	64.589	93.587	73.648	67.512
PFTD	88.725	90.678	73.923	90.254	91.553	74.519	92.618	93.433	76.945	95.024	96.345	79.863
PerFLID	85.609	86.432	92.604	87.300	87.641	93.600	89.347	89.427	95.007	91.965	92.127	98.004
CTGNN	80.214	87.945	89.567	82.563	88.230	90.873	84.553	90.351	92.678	87.254	93.784	92.861
VAEEVT	66.303	85.007	72.852	68.456	86.309	73.446	70.026	88.579	75.334	73.860	91.266	78.200
OSDN	64.817	93.612	97.331	65.981	94.113	98.522	66.321	96.452	100.212	68.006	99.819	103.475
BLIP	69.855	96.007	82.106	70.253	97.862	83.007	72.844	99.865	85.900	74.842	101.967	88.628
R-DFCIL	60.773	84.525	89.634	62.984	85.227	90.105	64.533	87.660	91.963	65.999	90.418	93.245
FACT	65.392	68.912	90.005	66.718	69.543	91.227	67.900	71.945	92.841	68.893	73.864	94.953
Our	59.674	70.200	63.776	60.245	70.765	64.859	60.878	71.249	65.762	61.246	72.993	66.847

Table 16
Latency analysis on different datasets.

Model	Dataset I/ (ms/sample)			Dataset II/ (ms/sample)			Dataset III/ (ms/sample)			Dataset IV/ (ms/sample)		
	5%	10%	15%	5%	10%	15%	5%	10%	15%	5%	10%	15%
TM-GAN	20.336	26.782	35.914	20.781	26.934	35.712	21.067	27.214	35.989	20.914	27.083	35.861
PFTD	23.875	30.143	39.276	24.163	30.298	39.154	24.428	30.587	39.463	24.296	30.462	39.338
PerFLID	31.259	37.904	47.138	31.842	38.017	46.908	32.114	38.296	47.152	31.967	38.152	47.023
CTGNN	18.601	24.367	33.452	18.954	25.127	33.264	19.236	25.381	33.547	19.108	25.274	33.412
VAEEVT	11.857	17.236	25.319	12.214	17.786	25.913	12.486	18.073	26.204	12.372	17.951	26.086
OSDN	22.025	28.918	37.087	22.487	28.643	37.512	22.764	28.918	37.786	22.653	28.809	37.675
BLIP	12.894	18.765	27.654	13.428	19.672	27.846	13.672	19.943	28.118	13.541	19.806	27.982
R-DFCIL	13.571	20.412	29.821	14.063	20.291	28.937	14.317	20.556	29.207	14.192	20.437	29.083
FACT	16.307	22.659	31.563	16.874	23.042	31.956	17.163	23.348	32.276	17.028	23.214	32.138
Our	8.624	11.631	15.872	9.012	12.041	16.118	9.268	12.314	16.427	9.137	12.189	16.302

set with the increasing number of categories, FAN-AFSIL achieves a favorable balance between discriminative performance and computational cost, making it highly effective in practical scenarios with limited data and gradually increasing categories. Prototype-based relational modeling does not result in prohibitive overhead, while still supporting incremental adaptation and open-set recognition efficiently. However,

for large-scale applications with significantly more attack categories, further optimization strategies such as prototype compression, sparse relational modeling, dynamic prototype selection, or hierarchical prototype organization may be required. Overall, these results confirm that FAN-AFSIL is practically feasible and well-suited for deployment in resource-constrained IoT environments.

5. Conclusions

In this study, we proposed FAN-AFSIL, a novel open-set IoT intrusion detection framework that integrates feature augmentation with adaptive few-shot incremental learning to address the challenges posed by fine-grained and data-scarce emerging attack categories. The proposed framework effectively enhances feature representation and decision boundary adaptation, enabling efficient learning under limited supervision. Extensive experiments conducted on four widely used IoT datasets demonstrate that FAN-AFSIL consistently outperforms existing baseline and state-of-the-art methods in open-set intrusion detection, while achieving favorable efficiency-accuracy trade-offs in terms of detection time and memory consumption. These results indicate its suitability for deployment in real-world, resource-constrained IoT engineering applications.

Despite the promising performance of FAN-AFSIL, several limitations still remain. The effectiveness of the proposed framework under extreme category overlap, severe class imbalance, and highly limited zero-day attack samples requires further investigation. Meanwhile, FAN-AFSIL can still enhance discriminative feature learning and adapt to emerging zero-day attacks under few-shot conditions, while initially identifying unseen attacks as unknown classes to reduce misclassification risk. Future work will focus on improving scalability and robustness through prototype optimization, active learning, generative sample synthesis, and cost-sensitive learning, as well as validating the framework on broader backbone architectures and diverse IoT environments.

CRediT authorship contribution statement

Wengang Ma: Writing – original draft, Validation, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Hekun Yang:** Validation, Software, Investigation, Formal analysis, Data curation. **Binbin Liu:** Visualization, Validation, Software, Formal analysis, Data curation, Conceptualization. **Junjiang He:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition, Conceptualization. **Xiaolong Lan:** Writing – review & editing, Visualization, Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Conceptualization. **Jiangchuan Chen:** Visualization, Validation, Supervision, Software, Investigation, Data curation. **Tao Li:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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